

Review Article

Recent developments in solid-solid phase change materials for thermal energy storage applications

Maoyong Zhi^{a,b,*}, Shan Yue^a, Lingling Zheng^a, Bingjian Su^a, Ju Fu^{a,b}, Qiang Sun^{a,b}^a College of Civil Aviation Safety Engineering, Civil Aviation Flight University of China, Guanghan 618307, Sichuan, China^b Civil Aircraft Fire Science and Safety Engineering Key Laboratory of Sichuan Province, Civil Aviation Flight University of China, Guanghan 618307, Sichuan, China

ARTICLE INFO

Keywords:

Solid-solid phase change materials

Energy storage

Preparation methods

Thermal management

ABSTRACT

Phase change materials (PCM) have been widely used in thermal energy storage fields. As a kind of important PCMs, solid-solid PCMs possess unique advantages of low subcooling, low volume expansion, good thermal stability, suitable latent heat, and thermal conductivity, and have attracted great attention in recent years. In this review, the application and preparation method of solid-solid PCMs were introduced. The preparation methods mainly consist chemical grafting, block polymerization and cross-linking polymerization. In terms of performance improvement, the enthalpy, thermal conductivity, mechanical properties, and thermal stability of PCMs were comprehensively illustrated. The research advances of solid-solid PCMs were mainly summarized in several application fields, including lithium-ion battery thermal management, solar energy storage, building energy conservation and other thermal storage fields. Finally, the current challenges and probable recommendations of solid-solid PCMs for thermal energy storage was put forward. Although some properties of solid-solid PCMs still need further research, they have good prospects for future thermal energy storage applications. This review provides latest progress and ideas for the design, preparation, and application research of new solid-solid PCMs.

1. Introduction

In recent years, natural resources and synthetic resources have faced a gradually depleted crisis, and these problems have seriously affected the sustainable utilization of resources [1]. Nowadays, most of traditional fossil fuels are not only non-renewable resources, but also they cause considerable pollution to the environment [2]. Every country is under obligation to make continuous efforts to develop new clean energy. Phase change materials (PCMs) for thermal energy storage have become one of good option for future clean energy. The phase change heat storage materials can store or release a large amount of heat during phase change process, and this latent heat enables it to maintain its own temperature constant [3]. Among all cooling technologies, PCMs are considered one of the most effective due to their low cost, simple structure, and high cooling efficiency [4]. Therefore, in recent years, PCMs have begun to receive attention from researchers in the fields of thermal energy storage and thermal management [5,6].

Thermal energy storage technology can improve thermal energy utilization efficiency, and it plays a key role in the development of renewable energy [7]. Among the three heat storage methods, including

sensible heat, latent heat, and chemical energy, latent heat storage technology has the unique advantages of high heat storage density and nearly isothermal working process, which has the broad application prospects in building energy conservation and many other fields. Fig. 1 illustrates a detailed classification of thermal energy storage materials [8].

PCMs have great potential in energy storage technology due to their high energy storage density and good isothermal properties [9]. They can store and release a large amount of energy during the phase change process. Therefore, it can be used to improve the efficiency of energy utilization, and meanwhile it can also reduce the negative impact on the environment [10]. PCM heat storage technology belongs to latent heat storage [11], and it can be classified as solid-solid, solid-liquid, gas-liquid, and solid-gas on the basis of the phase change characteristic. Due to the storage difficulty of gas, there are mainly solid-liquid PCMs and solid-solid PCMs in actual application [12].

Although the previous studies have paid more attention to solid-liquid PCMs than solid-solid PCMs, they have obvious shortcomings such as easy leakage, corrosiveness, and high packaging cost [13]. In comparison, solid-solid PCMs have the following advantages, such as no

* Corresponding author at: College of Civil Aviation Safety Engineering, Civil Aviation Flight University of China, Guanghan 618307, Sichuan, China.

E-mail address: zhimaoyong@cafuc.edu.cn (M. Zhi).

phase separation, small volume change, no leakage, non-corrosive, long service life [14]. Therefore, it meets the development needs of safety use and green environmental protection. In this review, the preparation methods of solid-solid PCMs were briefly outlined. Several important performance improvement studies of solid-solid PCMs were also reviewed. The recent research progress of solid-solid PCMs was summarized in lithium-ion battery, solar energy storage, building energy saving, and other thermal storage fields. The future development trends were also introduced.

2. Summary of PCMs

2.1. Definition of PCMs

As a well-known latent heat storage material, PCMs realize the storage and release of thermal energy during phase change process [15]. Because of their temperature within a certain range, PCMs are widely used in building energy conservation, electronic components, and lithium-ion batteries [16,17]. PCMs also have effectively prevent a series of thermal disaster issues. The latent heat of PCM is the heat of absorbed and released during vaporization, sublimation, and dissolution process [18–20]. Fig. 2 shows the working schematic diagram of phase change heat storage materials [21]. Latent heat storage is more attractive than sensible heat storage because of its ability to absorb and release heat at constant temperature. In addition, sensible heat storage has the disadvantages of low heat storage density. Therefore, latent heat storage of PCMs is one of the effective ways to store thermal energy because of its higher thermal energy storage ability.

2.2. Types and properties of PCMs

At constant temperature, PCMs can change from solid to liquid state by absorbing latent heat of melting and vice versa, or from liquid to gaseous state by absorbing latent heat of vaporization and vice versa, or from solid to gaseous state by absorbing latent heat of sublimation and vice versa [22]. According to the state of PCMs before and after phase change, it can be divided into solid-solid PCMs, solid-liquid PCMs, solid-gas PCMs and liquid-gas PCMs [23]. In practice, the application of solid-gas and liquid-gas PCMs is relatively difficult [24]. Solid-liquid PCMs is

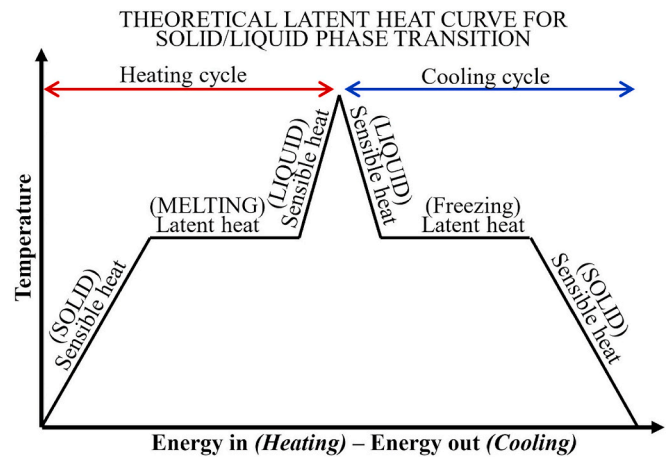


Fig. 2. Schematic representation of working of PCM [21].

one of the most widely used materials, but its leakage and corrosion problems are difficult to solve [25]. Solid-solid PCMs is in a solid state before and after phase change, and they have no leakage risk, small volume change, non-toxic and harmless, low requirements for containers, and long service life, thus they have become a research hotspot in recent studies [26,27].

For solid-solid PCMs, there are mainly two classification methods. According to phase change medium, it can be classified into inorganic salts, macromolecules, and polyols types. According to the material skeleton, it is classified into microcapsule, three-dimensional porous, nanotube, and polymer types [28,29]. According to the source of solid-solid PCMs, they can be divided into natural solid-solid PCMs and novel solid-solid PCMs. This review focuses on novel solid-solid PCMs. Nowadays, because PCMs can effectively alleviate the contradiction of imbalance of thermal energy supply and demand, they have been well applied in thermal energy storage, energy conservation and regulation, and other fields. Solid-solid PCMs can be used in lithium-ion batteries, building energy conservation, solar energy storage, and industrial waste heat recovery. In addition, the research on PCMs synthesized from renewable resources is gaining popularity.

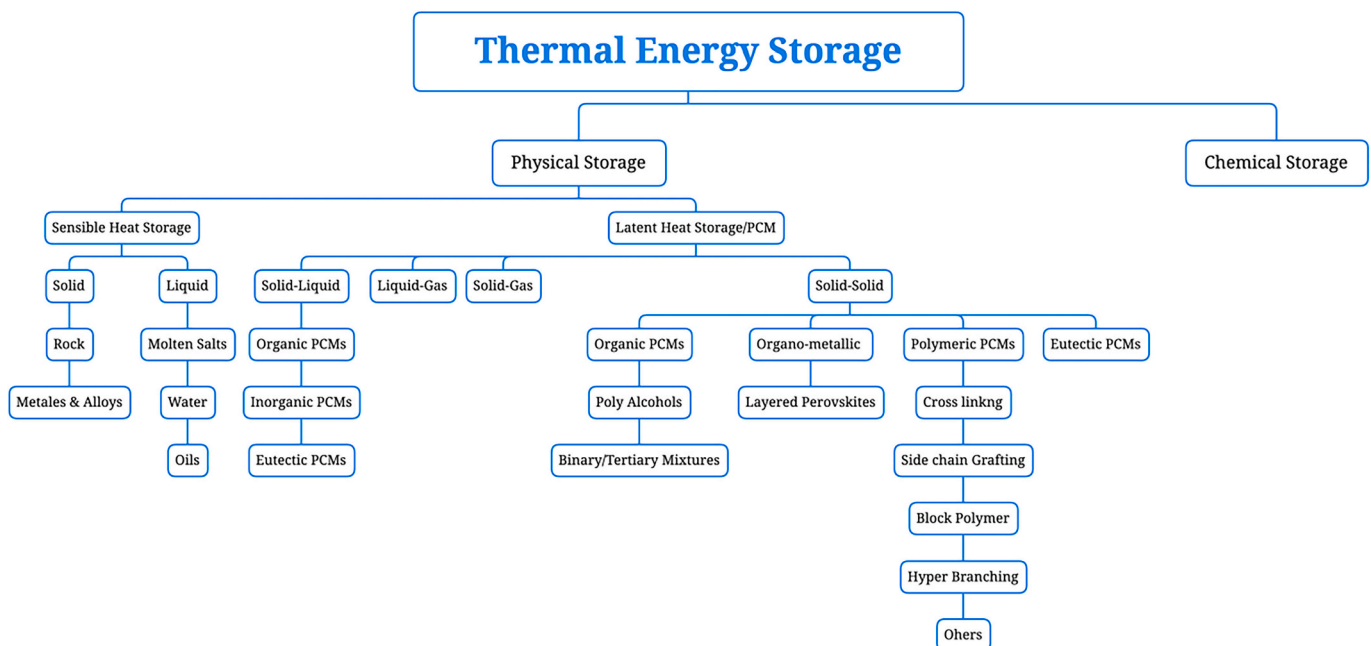


Fig. 1. Overview of thermal energy storage materials [8].

3. Overview of solid-solid PCMs

3.1. Preparation method of solid-solid PCMs

There are two main methods for preparing solid-solid PCM. The first is small molecules that change from one crystalline phase to another when heat is absorbed or released [30]. The solid-solid phase change of polyols is like that of inorganic hydrates, but polyols undergo at least one solid-solid phase change process before transforming into liquid [5]. Polyol organics are transformed from ordered crystal arrangements to disordered amorphous arrangements as the temperature increases, as shown in Fig. 3.

The second approach refers that the crystallizable components chemically bond into secondary structure to prevent free flow in an amorphous state [5]. This method mainly includes polymer solid-solid PCMs [31], and perovskites [32]. Composite solid-solid PCMs are composed of PCM and support material. PCM stores or releases heat when it melts and solidifies, while support material prevents the leakage of melting substance, keeping the system in solid state. Under special conditions, the stabilization strategy of support material is also a crucial aspect [33,34]. Several studies have demonstrated that suitable support materials can encapsulate maximum PCM content and achieve high thermal energy storage densities. The commonly preparation methods of solid-solid PCMs are summarized as follows.

3.1.1. Chemical grafting

The chemical grafting method refers that PCMs are obtained by using the functional groups of framework material to react with phase change macromolecules or monomers [35,36]. As shown in Fig. 4, the phase change temperature is tailored by adjusting the chain length of soft segments or rigidity of polymeric backbone [5]. Common skeleton materials include cellulose [37], polystyrene (PS) [38], etc.

Guo et al. synthesized a solid-solid PCM by aqueous phase precipitation polymerization, using the amphiphilic polyethylene glycol-maleic acid monoester macromonomer as side chain and the polyacrylonitrile (PAN) as a skeleton. The melting enthalpy and crystallization enthalpy of grafted polymer reached 74.16 J/g and 65.40 J/g respectively [39]. Sari et al. synthesized a series of polyethylene glycol (PEG) grafted isocyanate-terminated polymers using three different molecular weights. Thermal conductivities of solid-solid PCMs were in a range of about 0.18 to 0.19 W/(m·K). Moreover, the PCMs were durable thermally until about 300 °C, and they showed good thermal reliability even after 5000 cycles [40].

Han et al. prepared a solid-solid PCMs using cellulose as the skeleton and polyoxyethylene (2) hexadecyl ether (E2C16) as side chain. The enthalpy and degree of substitution of this graft polymer could be adjusted by E2C16 content, reaction temperatures, and after-treatment process [41]. Mu et al. successfully synthesized a series of poly

(styrene-co-acrylonitrile) grafting with palmitic acid copolymers (SAN-g-PA) solid-solid PCMs from poly(styrene-co-acrylonitrile) (SAN) and palmitic acid (PA). The results showed that PA molecule was grafted onto the SAN, and the crystalline morphology and crystal structures of SAN-g-PA were different from palmitic acid and SAN. Besides, SAN-g-PA possessed suitable phase change temperature, high enthalpy, and good thermal stability [42].

Sari et al. synthesized poly (styrene-co-maleic anhydride)-graft-fatty acids (SMA-g-FA) copolymers solid-solid PCMs for thermal energy storage. SMA-g-FA copolymers had a phase change temperature of about 40–60 °C, and latent heat in range of 84–127 J/g. The solid-solid PCMs had good latent heat storage ability even after 5000 thermal cycling. Moreover, their thermal conductivity values were ranging from 0.15 to 0.19 W/(m·K) [43].

Han et al. synthesized a series of cellulose-g-polyoxymethylene (2) hexadecyl ether (cellulose-g-E6C16) copolymers using toluene 2,4-diisocyanate as a coupling reagent. Its heat storage ability and phase change temperature depended on the degree of substitution. The crystallization behavior of grafted E6C16 was not affected by cellulosic backbone. Compared with E6C16, cellulose-g-E6C16 showed better thermal stability [44]. Huang et al. synthesized a series of copolymers PCMs by grafting methoxypoly (ethylene glycol) to the main chain of poly (acrylonitrile-co-allyl glycidyl ether), in which PAA was the skeleton, and MPEG was a functional side chain. They stored and released heat during the phase change process [45].

Sari et al. synthesized poly(styrene-co-maleic anhydride)-graft-polyethylene glycol copolymers, which exerted heat storage ability by the phase change from crystalline phase to amorphous phase of PEG. The copolymers showed solid-solid phase change behavior between 40 °C–45 °C and considerable latent heat capacity of 107–155 J/g [46]. Cheng et al. prepared polyethylene glycol-polyacrylamide graft copolymer PCM (PPGC-PCM). PPGC-PCM improved the moisture resistance and thermal insulation properties of asphalt mixture. As the maximum nominal particle size of asphalt mixture decreased, PPGC-PCM had more significant impact on its mechanical properties [47].

Gu et al. prepared PEG/MMT solid-solid PCMs by graft copolymerization method using montmorillonite (MMT) as skeletons. The influence of different molecular weight PEG on the capability and structure of phase change latent heat and crystallization behavior were investigated. The PEG/MMT solid-solid PCMs could improve enthalpy value and thermal stability [48]. Mu et al. synthesized a novel poly(styrene-co-acrylonitrile)-graft-polyethylene glycol (SAN-g-PEG) copolymer by grafting PEG into the main chain of poly(styrene-co-acrylonitrile). SAN-g-PEG had the phase change temperatures in the range of 23–368 °C and the latent heat enthalpy between 66.8 and 68.3 kJ/kg [49].

Guo et al. prepared a novel PCM from polyethylene glycol acrylic acid (PEGA) and poly (decyl glycerol acrylate) (PDGA) using a free radical solution polymerization method. The crystallization enthalpy of

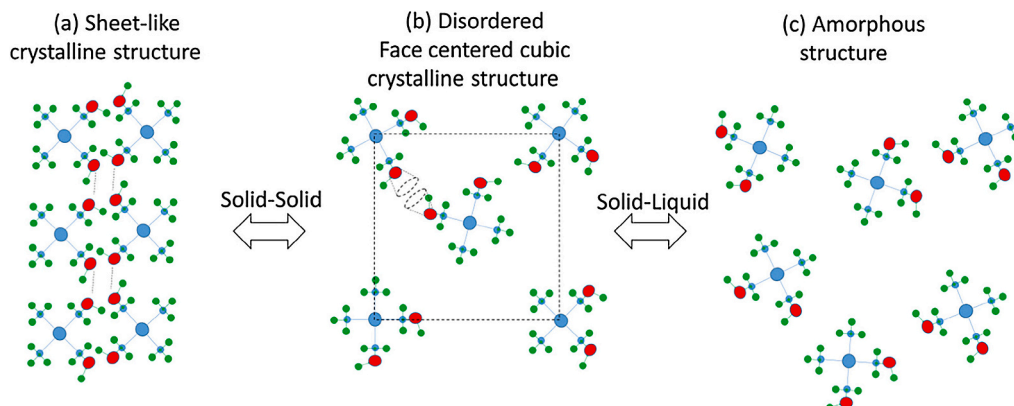


Fig. 3. Schematic diagram of polyalcohol type solid-solid PCMs: (a) tetrahedral configuration; (b) face centered cubic; (c) disordered amorphous structure [5].

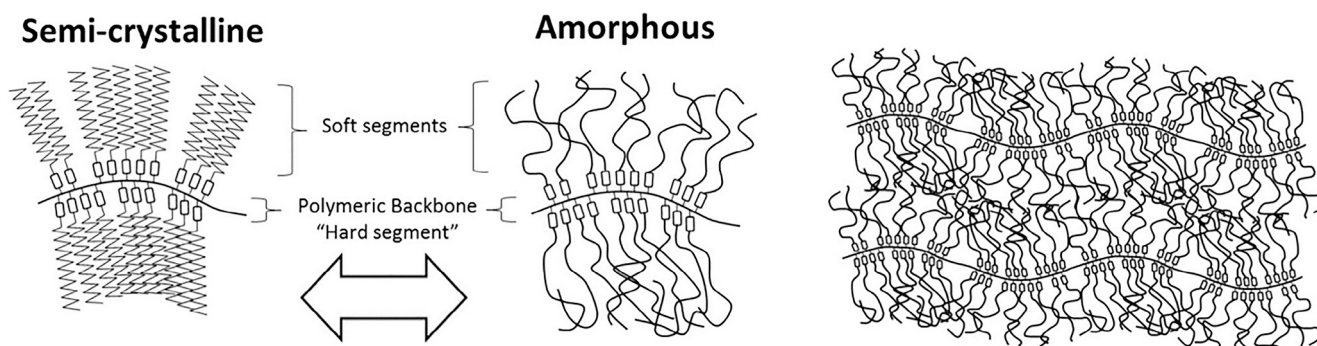


Fig. 4. Schematic diagram of polymeric graft solid-solid PCMs and their phase change [5].

PCM reached 141.3 J/g, and it had good thermal stability [50]. Sari et al. used polystyrene (EPS) grafted palmitic acid (PA) to prepare a new solid PCM, and the PCM exhibited good energy storage density, thermal stability, and cycling stability [51]. Next, Sari et al. prepared a solid-solid PCM by grafting styrene-co-allyl alcohol with stearic acid (SA). Compared with polystyrene graft palmitic acid copolymers, the thermal stability and heat release performance was significantly improved [52]. Li et al. prepared solid-solid PCMs using chemical grafting of PEG and cellulose, and the PCM had high thermal storage density and good thermal stability [53].

The grafted structures can regulate the functions of final products according to different substances. As the temperature rises to the phase change temperature of side chain, heat is stored through the phase change endothermic mechanism. At the same time, the fluidity of soft segment is restricted by hard segment, so that the polymer remains solid state [54]. The relevant performance is shown in Table 1.

3.1.2. Block copolymerization

The solid-solid PCM prepared by block copolymerization method is a linear polymer by alternately polymerizing chain segments. In the process of block copolymer copolymerization, the flexible polymer and crystalline layers are alternately polymerized [55]. Many linear polymers undergo reversible liquid-solid phase changes during cooling process. As shown in Fig. 5, during the crystallization process, the flexible polymeric chains align into crystalline lamellar plates because of the difference of cooling rate [5]. Over the past years, the relevant researches are summarized in Table 2.

Huang et al. synthesized a thermally-induced shape memory multi-block poly(ether-ester)s (PBSEGs) by two-step transesterification reaction. The copolymers displayed good hydrophilic and excellent shape-memory properties [56]. Pielichowska et al. modified polyurethane-based PCMs (PUPEG) with graphene using one-step bulk

polymerization method. PUPEG without chain extender exhibited higher latent heat and better thermal stability [57]. Wu et al. prepared a kind of flexible PCM with well-retained flexibility and wide temperature range. Because of the flexibility and high thermal conductivity of 1.64 W/(m·K), the flexible PCM has high latent heat of 102 J/g, suitable phase change temperature range of 40–50 °C, and showed outstanding battery thermal management performance [58]. Tian et al. prepared solid-solid PCMs by introducing π - π stacking to form physical cross-linking points in linear PEG chains. The strong intermolecular forces not only prevented the leakage of PEG even at 130 °C but also endowed PCMs with good heat resistance and high toughness (172.44 MJ/m³). Besides, the addition of CNT could further enhance the photo-thermal conversion ability and thermal conductivity of PCMs [59]. Jiang et al. prepared a novel flexible solid-solid PCMs by incorporating polyethylene glycol 8000 and 2,6-dimethyl piperazine into polymeric skeleton. The flexible PCMs showed excellent shape stability performance and solvent resistance, and had the highest latent heat of 106.3 J/g [60]. By chemically bonding different repeating units, many new linear, cyclic, and branched molecular structures continuously emerge [54].

3.1.3. Cross-linking polymerization

Cross-linking polymerization is another method to synthesize solid-solid PCMs with nonlinear structures [61]. The linear copolymer and backbone serve as the connection points between molecular chains to create a leak-proof 3-dimensional copolymer network [62]. The solid-solid PCMs with cross-linked network structure has a stable shape.

Wang et al. prepared a PEG-containing solid-solid PCM based on dynamic ferric ion cross-linking, which was in solid state even at 90 °C. This PCM showed excellent thermal energy storage capacity, and its chemical structure and thermal properties are almost unchanged after reprocess [63]. Zhao et al. successfully synthesized water insensitive and biodegradable PEG-based solid-solid PCMs using trimethylolpropane

Table 1

Summary on thermal properties of solid-solid PCMs prepared by chemical grafting.

Author	Ref.	Composite PCM	Melting temperature (°C)	Crystallization temperature (°C)	Latent heat (J/g)	Thermal conductivity (W/(m·K))	Thermal stability (°C)
Guo et al.	[39]	MA + PEG + AN	51.45	16.27	74.16	0.066	335
Sari et al.	[40]	PEG + TDI + PSAA	53.93	40.42	121.98	0.19	345
Mu et al.	[42]	SAN + PA	34.5	31.3	24.4	–	392
Sari et al.	[43]	SMA + FA	61	55.7	117.8	0.15–0.19	280
Han et al.	[44]	E2C16 + TDI + Cellulose	28.6	21.6	33	–	294
Huang et al.	[45]	PAA + MDI + MPEG	53.66	45.02	84.89	–	288
Gu et al.	[46]	PEG + TDI + MMT	40.62	–	79.69	–	348.5
Mu et al.	[47]	SAN + PEG	36.8	26.8	68.24	–	120
Guo et al.	[48]	PEGA + PDGA	38.97	57.58	141.3	–	210
Sari et al.	[51]	PA + EPS	19.18	18.72	39.78	–	160
Sari et al.	[52]	SA + EPS	29	30	71	–	321
Li et al.	[53]	Cellulose + PEG	50.4	28.9	149.1	–	250

Note: T_{exo} refers to exothermic phase change temperatures; T_{endo} refers to endothermic phase change temperatures.

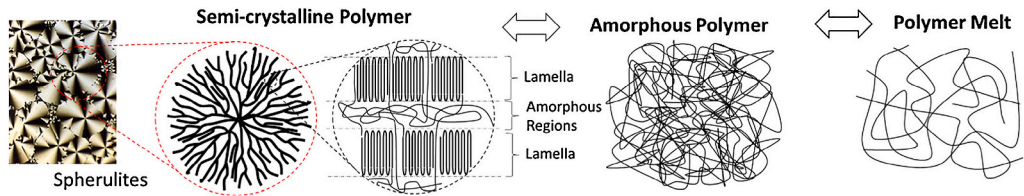


Fig. 5. Schematic diagram of semi-crystalline solid-solid PCM [5].

Table 2
Summary on thermal properties of solid-solid PCMs prepared by block copolymerization.

Author	Ref.	Composite PCM	Melting temperature (°C)	Crystallization temperature (°C)	Latent heat (J/g)	Thermal stability (°C)
Huang et al.	[56]	PEG + BDO + C ₆ H ₆ O ₄	112.4	69.52	53.97	261
Pielichowska et al.	[57]	PEG + MDI + BDO	59.4	57.6	140.8	265
Wu et al.	[58]	TPC-et + EG + PA	40–50	–	101104	–
Tian et al.	[59]	PEG + MDI + BP	58.2	36.7	105.9	300
Jiang et al.	[60]	IPDI + DPA + THDI	49.2	27.2	106.3	–

tris(1-aziridine propionate) (TMPTAP) as cross-linking agent by solvent-free bulk polymerization. PCMs had the similar crystalline structure to PEG, while the crystal size of PCMs was smaller than PEG. The phase change temperature of PCMs was in the range of 25–65 °C and the latent heat was about 103–108 J/g, respectively. Moreover, the supercooling degree of PCMs decreased [64].

In order to improve the thermal conductivity and thermal stability of PCMs, Xia et al. designed a novel polymeric solid-solid PCMs with lamellar structures by using PEG as the phase change ingredient, 4,4'-diphenylmethane diisocyanate (MDI) as the cross-linking agent, and graphene oxide (GO) as the skeleton material [10]. The schematic illustration for polyurethane prepolymer is shown in Fig. 6(a), and

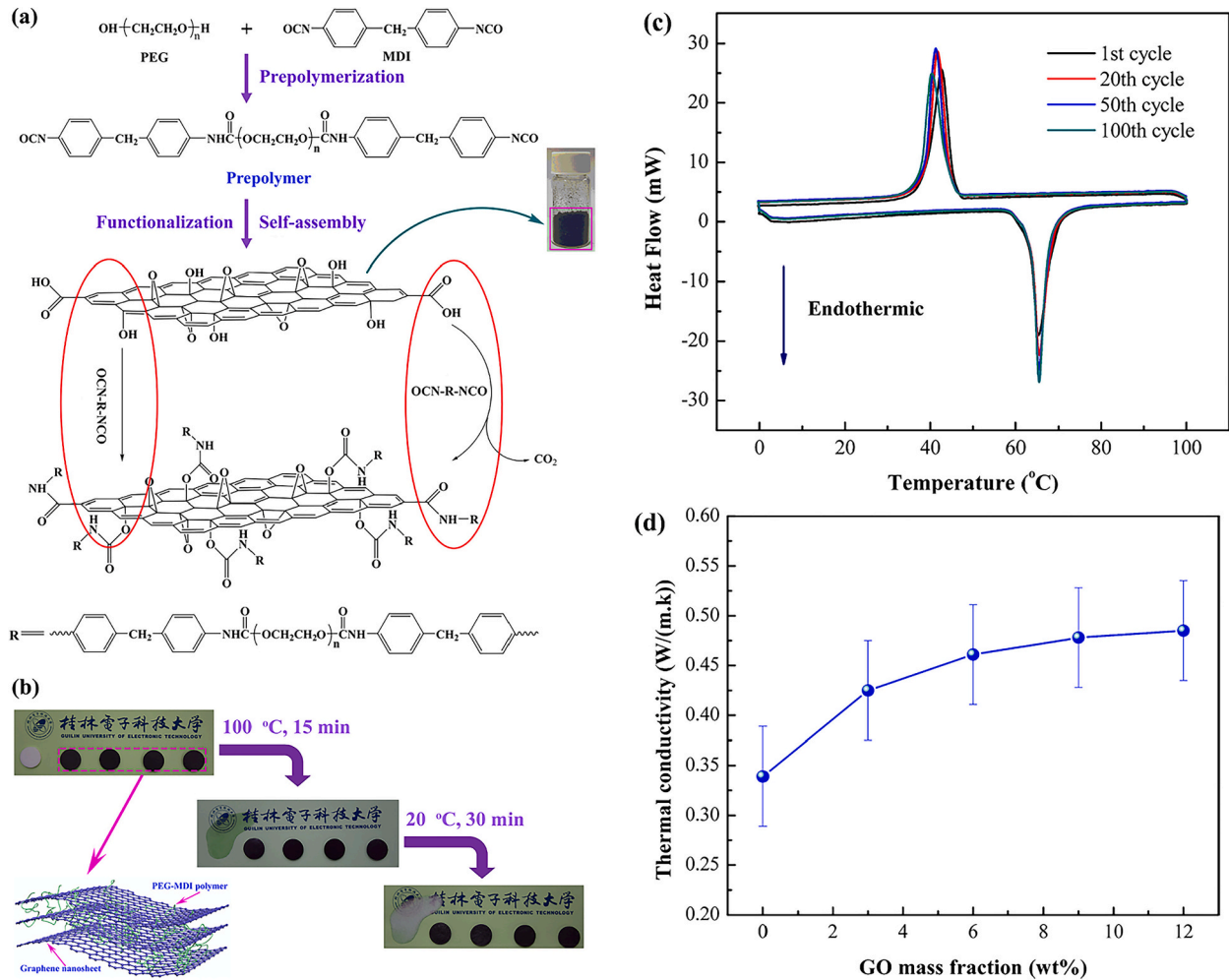


Fig. 6. (a) Schematic illustration for polyurethane prepolymer, (b) grafted-polymerization and self-assembly process, (c) DSC curves of PCM-3 over different thermal cycles, and (d) thermal conductivity [10].

grafted-polymerization and self-assembly process is displayed in Fig. 6 (b). GO nanosheets can enhance the thermal conductivity of solid-solid PCMs. As shown in Fig. 6(c) and (d), this molecular structure endows the solid-solid PCMs with excellent phase change behavior, good thermal cycling stability, and high thermal conductivity [8]. As a result, the solid-solid PCMs display a fast thermal response rate, and outstanding thermal regulating performance, which can maintain their temperature in the range of 50–57 °C for about 410 s.

Harlé et al. synthesized linear PU (PUL) and cross-linked PU (PUX) using hexamethylene diisocyanate (HMDI), poly (ethylene glycol) (PEG) and glycerol. Three parameters were varied in the synthesis of PUs, such as PEG molecular weight, stoichiometric, and cross-linker content. The solid-solid PCMs showed latent heat between 79 J/g and 108 J/g and the phase change temperature range of 22–50 °C [65]. Lin et al. synthesized a form-stable, recyclable solid-solid PCMs from thermal reversible dynamic Diels-Alder bonds. The solid-solid PCMs exhibited excellent shape, heat storage stability, and reprocessing ability. Moreover, the halloysite nanotubes decorated with Fe₃O₄ nanoparticles improved the thermal conductivity [66].

Du et al. synthesized tri-maleimide end-capped cyclotriphosphazene flame retardant (TMCTP) via the nucleophilic substitution reaction of 1,3,5,2,4,6-triazatriphosphorine-2,2,4,4,6,6-hexachloride and N-(2-hydroxyethyl) maleimide. The cross-linked PCMs were fabricated by bonding PEG and TMCTP skeleton through furan/maleimide Diels-Alder reaction, which displayed excellent self-healing and recycling, and good fire safety [67]. The synthetic solid-solid PCMs by cross-linking has a relatively stable network structure, which improves the heat storage capacity of PCM. The relevant performance is shown in Table 3. Huang et al. fabricated a dynamic urethane bond cross-linked solid-solid PCM with PEG as phase change functional segments which was in solid state even at 130 °C. Meantime, it exhibited excellent solid-state plasticity due to dynamic urethane bonds. The crystalline PEG endowed this solid-solid PCM with high enthalpy value of 85 J/g, remarkable mechanical flexibility (ultra-stretchability of 600 %), thermal reliability and chemical stability [68]. Chen et al. provided a new idea to prepare form-stable PCMs with high phase change enthalpies. Glucose and PEG was respectively used as skeleton and phase change chain. The form-stable PCMs displayed high thermal storage density, and their enthalpy efficiency were much higher than that of the traditional form-stable PCMs. Moreover, the form-stable PCMs have excellent thermal reliability and reusability [69].

Wang et al. synthesized a novel solid-solid PCMs by two-step condensation reaction of PEG, neopentyl glycol and 4,4'-diphenylmethane diisocyanate. The results indicated that the phase change enthalpy and phase change temperature respectively reached to 120.45 J/g and 37.32 °C respectively [70]. Fu et al. synthesized the polyester-based solid-solid PCMs via ionic crosslinking reaction between carboxyl-functionalized PCMs and zinc cations. The PCMs were synthesized by polyesterification reaction between pyro-mellitic dianhydride and diol compound. PCMs had an initial decomposition temperature above 300 °C and reprocess-ability/recyclability at 120 °C [71].

To sum up, solid-solid PCMs can be prepared by grafting, blocking, and cross-linking polymerization methods. Use solid-liquid PCMs with

high phase change enthalpy and appropriate phase change temperature as soft chains. When the temperature reaches the melting point of the “soft segment”, a solid-liquid change occurs in the “soft segment”, which absorbs heat through phase change. Due to the combination of soft segments and hard segments through chemical bonding, the flow of liquefied soft segments can only be limited by the skeleton structure, resulting in a macroscopic solid-solid phase change. The temperature and enthalpy of solid-solid PCMs are regulated by the structure and chain length of soft segments and main chains. The higher the proportion of soft segments in the polymer chains, the poorer the thermal stability, the lower the phase change temperature, and the greater the enthalpy. Compared with block copolymers and graft copolymers, cross-linked polymers have higher thermal stability. The preparation of cross-linked solid-solid PCMs with high enthalpy is worth studying.

3.2. Several important performances of solid-solid PCMs

3.2.1. Thermal conductivity

Although PCMs have strong heat storage capacity, the premise is that heat can be quickly transferred inside the PCM, avoiding uneven phase change that leads to low heat storage. The thermal conductivities of solid-solid PCMs are often low, which affects the actual usage. Therefore, it is necessary to enhance the thermal conductivity of solid-solid PCMs [72,73].

Wang et al. prepared composite PCM with inorganic-organic hybrid flame retardant as supporting material and PEG as working materials. When the content of PEG was 90.4 wt%, and the phase change temperature and latent heat of composite PCM changed weakly, the thermal conductivity increased by 111 % compared with that of pure PEG. Furthermore, the peak heat release rate of composite PCM reduced by 33.4 % and the burning time delayed 36 s. Furthermore, Fig. 7 shows the SEM images and DSC results [74].

Elsayed et al. investigated heat transfer characteristics and thermal storage performance of enhanced neopentyl-glycol (NPG) [75]. The heat storage of NPG composite containing 6 % SiO₂ and 6 % Al was more than the composite containing 6 % Al and 6 % Cu. Also, compared to the other compositions, the best heat storage property was achieved at the addition of 6 % Al + 3 % SiO₂ + 3 % TiO₂. Luo et al. obtained composite PCMs with high photothermal conversion performance using pentaerythritol (PE) as a phase change heat storage medium and gallic acid modified graphene nanosheets (GNPs) as light absorbers and thermally conductive filler, as shown in Fig. 8(a). The preparation process is shown in Fig. 8(b). The composite PCMs containing 15 wt% GNPs had the highest photothermal conversion efficiency, reaching 93.86 %, which was 247.76 % higher than that of PE, and the latent heat of phase change was 190.88 J/g. Compared with thermal conductivity PE of 1.02 W/(m·K), the thermal conductivity of PCMs containing 15 wt% GNPs reached 3.86 W/(m·K) [76]. The photothermal conversion test system is displayed in Fig. 8(c). Besides, the photothermal conversion characteristics of PE and PCMs were also investigated under simulated sunlight.

Mani et al. prepared copper-pentaerythritol composites using copper nanoparticles as an additive of pentaerythritol [77]. The high thermal conductivity of copper improved the heat transfer of thermal storage systems. The results indicated that the thermal conductivity of

Table 3
Summary on thermal properties of solid-solid PCMs prepared by cross-linking polymerization.

Author	Ref.	Composite PCM	T _{endo} (°C)	T _{exo} (°C)	Latent heat (J/g)	Thermal conductivity (W/(m·K))	Thermal stability (°C)
Harlé et al.	[65]	PEG + HMDI + glycerol	51.7	39.5	108	0.231	280
Lin et al.	[66]	PEG + IPDI + EKN	48.84	18.09	116.8	0.223	250
Du et al.	[67]	PEG + MDI + BDO + furfuryl alcohol	44.7	27.3	88.1	–	200
Huang et al.	[68]	PEG + MDI + TEOA	49.12	39.89	87.83	–	261
Chen et al.	[69]	PEG + MDI + glucose	56.71	46.13	108.7	–	388.6
Wang et al.	[70]	PEG + MDI + NPG	37.32	–	120.45	–	400.1
Fu et al.	[71]	PEG + PMDA + EG + TEA/Zn(OAc) ₂	60	42	120	–	350

Note: T_{exo} refers to exothermic phase change temperatures; T_{endo} refers to endothermic phase change temperatures.

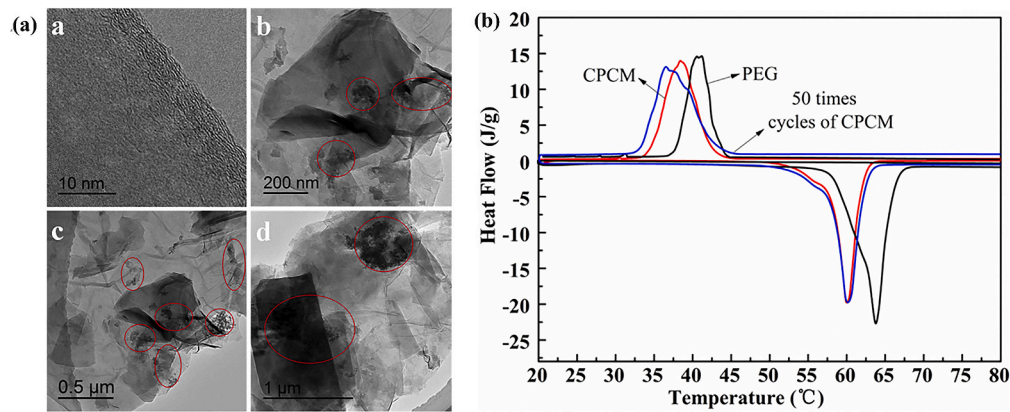


Fig. 7. (a) SEM images of GO-ADCT-NH₂, and (b) DSC curves of PEG and composite PCM [74].

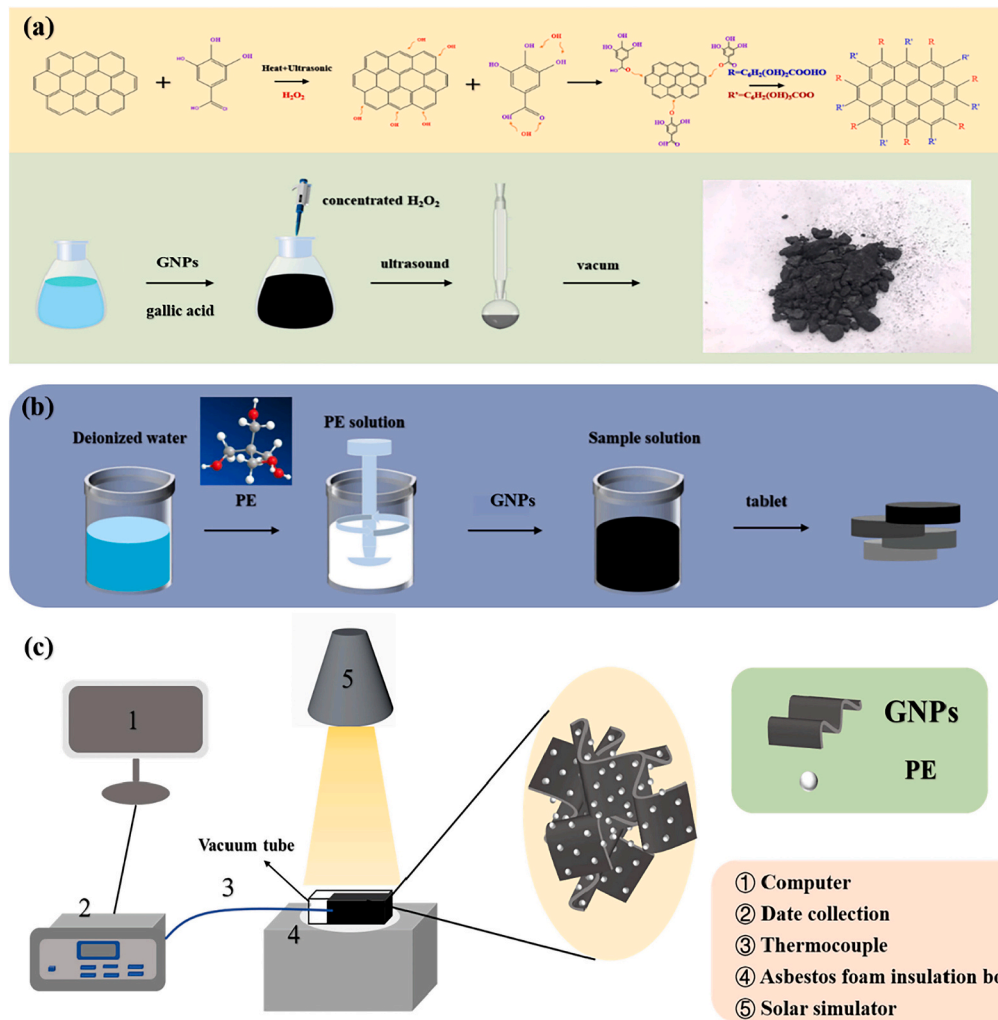


Fig. 8. (a) Modification of GNPs, (b) synthesis of composite PCMs, and (c) photothermal conversion test diagram [76].

pentaerythritol and copper-pentaerythritol composites decreased with temperature. Fig. 9 shows the testing equipment. The enthalpy changes of composites were not different from that of pentaerythritol. However, the composites absorbed lower energy for tetragonal to cubic transition.

Zhang et al. prepared a high-performance thermal interface material with solid-solid phase change properties by using reduced graphene oxide (rGO)-coated polyurethane (PU) foam as a filler, and segmented polyurethane (SPU) as a matrix [78]. The pure SPU and rGO/SPU

composite exhibited proper phase change temperature, high latent heat, and no leakage. The thermal conductivity of rGO/SPU composite was 63 % higher than that of pure SPU at an ultra-low rGO content of 0.8 wt %. Besides, SPU displayed better heat transfer performance than the PU without phase change properties. Generally, carbon materials can improve thermal conductivity of PCMs. The complete physical network structure helps the high thermal conductivity, which also improve the overall latent heat utilization rate [68].

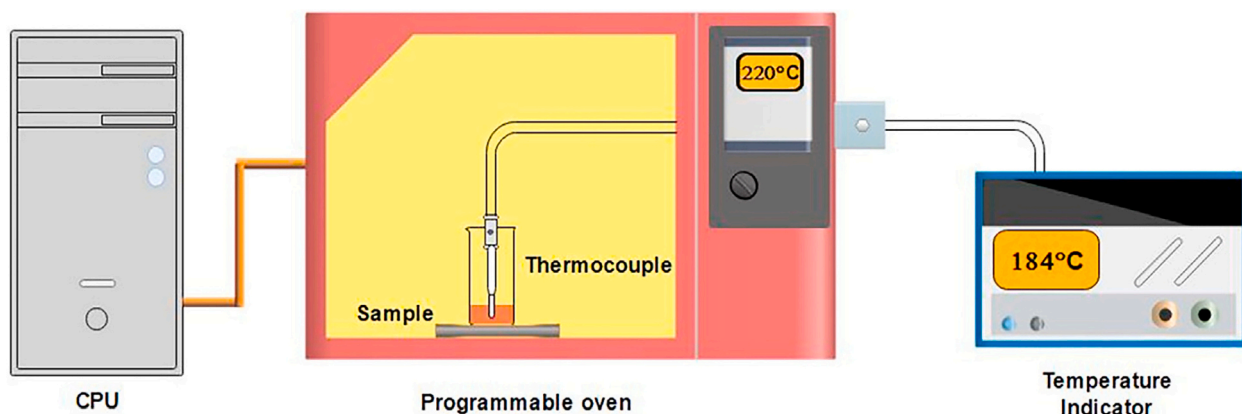


Fig. 9. Schematic diagram of the experimental platform for heat transfer tests [77].

3.2.2. Phase change enthalpy

The addition of nanoparticles can improve the thermal conductivity, and it also leads to a decrease in the melting enthalpy. Actually, thermal conductivity and melting enthalpy are two important parameters of PCM. The thermal conductivity affects the heat storage rate, and the melting enthalpy determines the heat storage density. Fig. 10 shows the initial model of composite PCM containing nanoparticles [79].

Li et al. fabricated melamine/formaldehyde/polyethylene glycols (MFPEG) through amine-aldehyde condensation and aldolization reaction [80]. MFPEGs had the similar crystal structure to PEG, and the maximal latent heats were respectively 109.4 J/g and 103.9 J/g during

heating and cooling cycles. The decomposition temperature of MFPEGs was about 340–455 °C at nitrogen atmosphere. Li et al. prepared the lignin-based PCMs using PEG and porous carbon material by vacuum impregnation. The porous carbon has the largest loading capacity as much as 85 % of PEG, and the largest latent heat of PCMs was 89.7 J/g [81].

Alkan et al. prepared a series of PUs via the condensation reaction of PEGs and diisocyanates with different molecular weights. The produced PUs had typical solid-solid phase change properties with high enthalpies of 179 J/g and phase change points of 19–60 °C. The copolymers could be used for thermal energy storage and temperature control fields [82].

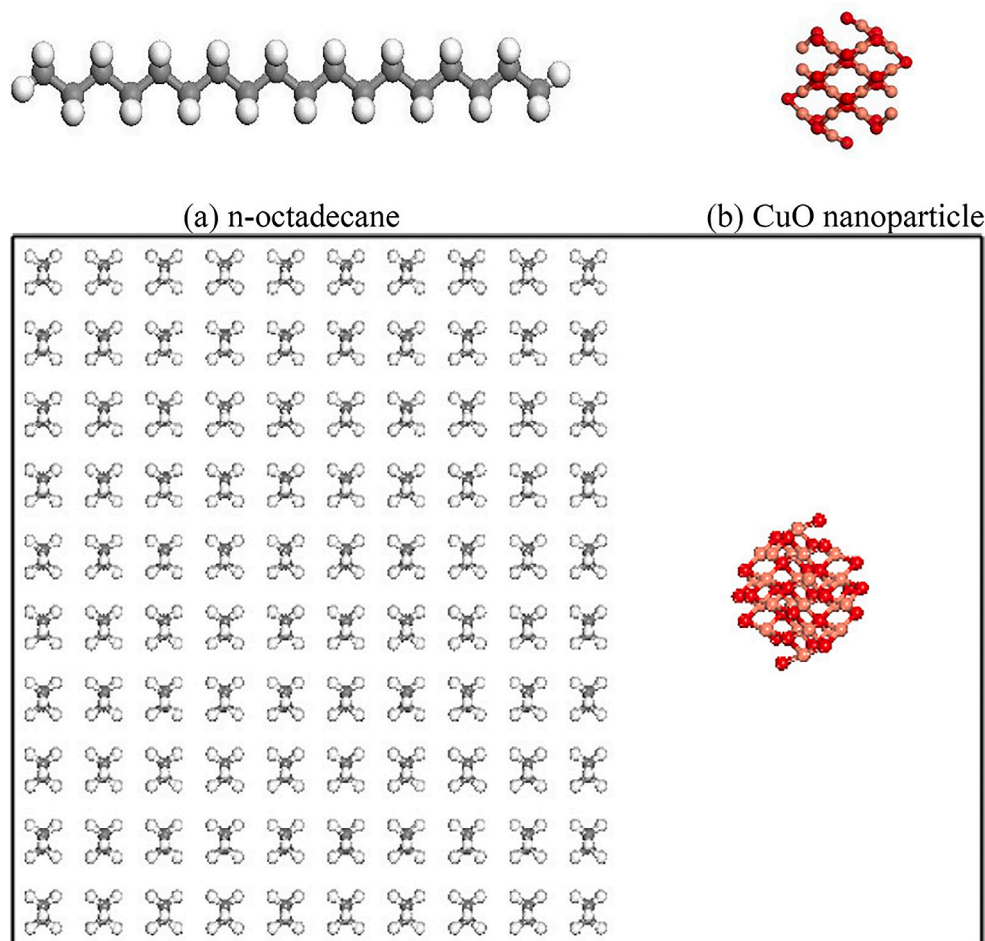


Fig. 10. Initial structure of composite PCM containing nanoparticles [79].

Zhou et al. prepared multifunctional polyurethane-based PCMs through facile impregnation method [83]. The schematic diagram of the preparation is shown in Fig. 11(a). Zinc oxide (ZnO) and graphene aerogel (GA) improved the latent heat of the system from 86.8 to 108.1 J/g. Furthermore, with help of light absorption of ZnO and GA, as well as the electrical conductivity of GA, the photo-thermal and electro-thermal energy conversion efficiency were respectively 80.1 % and 84.4 %. Fig. 11(b) displays the schematic diagram of conversion and storage mechanisms of photothermal and electrothermal processes.

Du et al. synthesized a series of novel *comb*-polyurethane PCMs (CP-PCMs) through the reaction between methoxypolyethylene glycol and 1,4-butanediol. CP-PCM possessed optimal phase change property using isophorone diisocyanate or 1,6-hexamethylene diisocyanate as diisocyanate and MPEG with molecular weight of 5000 as soft segment. In addition, the phase change enthalpy and temperature of CP-PCMs decreased as the amount of MPEG decreased [84].

The melting enthalpy is an important factor for thermal energy storage, but the enthalpy of solid-solid PCMs is often low. Thermal energy storage is an integral part of many energy systems and an important tool for achieving carbon neutrality. Among PCMs, inorganic hydrated salts and low melting point metals have a high phase change latent heat, so the improvement of melting enthalpy of organic solid-solid PCMs needs to be explored in depth [85].

3.2.3. Thermal stability

The successful utilization of latent heat energy storage systems largely depends on the thermal reliability of PCMs [86]. The economic feasibility of PCMs depends on the lifetime and cost of materials. Although the fabrication, leakage, thermal conductivity, and heat storage properties of PCMs have been extensively studied, there is insufficient research on the thermal stability of PCMs [87].

Raj et al. synthesized a new class of manganese based layered perovskite solid-solid PCMs by a liquid phase reaction method, and the effect of thermal cycling was compared at 0, 500, and 1000 cycles. The procedure is shown in Fig. 12(a) and (b). The solid-solid PCMs displayed thermal and chemical stability after around 1000 cycles due to the presence of metal substrate [88].

Kong et al. prepared thermosetting polyurethane PCMs through solvent-free bulk polymerization from PEG and polyaryl polymethylene isocyanate. In melting and freezing process, the maximum latent heat of PCMs respectively reached 111.7 J/g and 110.4 J/g. Besides, the prepared PCMs showed a good thermal stability with its onset degradation temperature up to 360 °C [89]. Yang et al. prepared novel polymeric solid-solid PCMs through solvent-free bulk polymerization using employing PEG as phase change functional segments, diphenylmethane diisocyanate as coupling agent and xylitol as curing agent. Compared with pure PEGs, the solid-solid PCMs showed the same crystalline structure and smaller crystal size. Meanwhile, the PCMs possessed a latent heat exceed 60 J/g and phase change temperature in the range of

20–45 °C. Their onset decomposition temperatures were above 300 °C [90].

Liu et al. prepared two components-based PEG solid-solid PCMs via free-radical polymerization. One of the components consisted of PEG-based polyurethane acrylate prepolymer and extra PEG. The prepared PCMs displayed similar crystalline structures to that of PEG. And the form stable PCMs had higher latent heats than that of traditional PCMs. Moreover, the prepared PCMs exhibited good thermal reliability and reusability in a long-time usage [91]. Chen et al. synthesized three kinds of new polymeric solid-solid PCMs with different crosslinking structures for thermal energy storage. In solid-solid PCMs, three hexahydroxy compounds (sorbitol, dipentaerythritol and inositol) were individually employed as the molecular skeleton, and PEG was used as phase change functional chain. The prepared solid-solid PCMs possessed an applicable temperature range of 30–70 °C, and the phase change enthalpy was up to respectively 107.5 J/g and 102.9 J/g during heating and cooling process. Moreover, they had good reusability and thermal reliability [92].

The lifetime of PCMs depends on their thermal cycling ability after repeated use. After many studies, polyethylene glycol has good thermal stability after many thermal cycles. Therefore, it can be used as a base material to improve the thermal stability of solid-solid PCMs [93]. The introduction of redox initiation systems and cross-linked networks could increase the crosslinking density and change the molecular chain structure of solid-solid polymeric PCMs, and its thermal stability can be improved to a certain extent. By utilizing the specific characteristic of PCMs that absorb or release heat during phase change process, it is very effective to regulate the battery temperature within a reasonable range. The larger latent heats of PCMs, the more heat they absorb in process of battery thermal management. The higher thermal conductivity, the better heat transfer effect within PCMs, which helps to fully utilize the endothermic effect of PCMs.

4. Research advances of solid-solid PCMs in thermal energy storage applications

In recent papers, the phase change points of solid-solid PCMs could be selected in a wide temperature range of −5 °C to 190 °C, which is suitable to be applied in many fields, such as lithium-ion batteries, solar energy, build energy conservation, and other thermal storage fields [94]. Therefore, solid-solid PCMs have broad application prospects. The great potential applied in thermal management system depends on the advantages of solid-solid PCMs, including good thermal conductivity, high storage density, low subcooling degree, no leakage risks and so on. The main application fields of solid-solid PCMs include lithium-ion battery, solar energy storage, building material and so on.

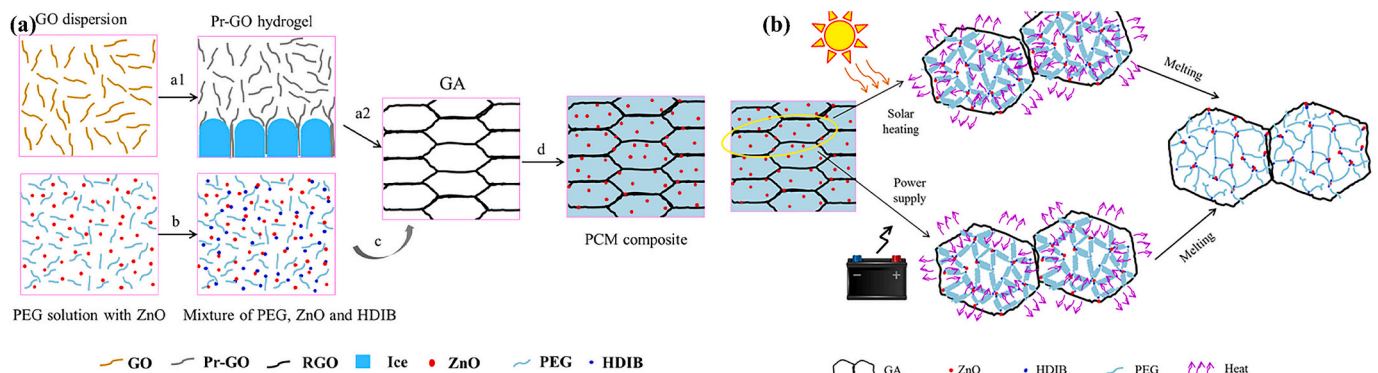


Fig. 11. (a) Fabrication procedure of PCM, and (b) light-thermal and electro-thermal conversion and storage mechanism [83].

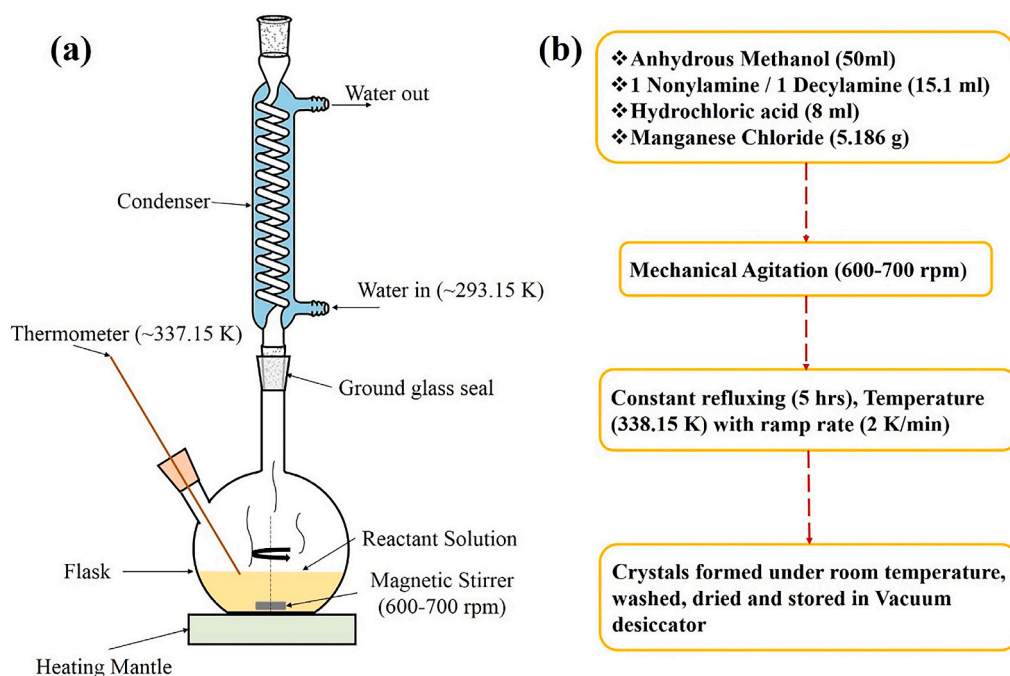


Fig. 12. (a) Experimental equipment, and (b) preparation steps of layered perovskite [88].

4.1. Lithium-ion battery

Over the past few decades, fossil fuels cause the severe environmental problems. As a very important energy storage technology, lithium-ion batteries have been widely used in electrical equipment, electric vehicles, power systems and robotics, and aviation due to the advantages of high power and energy density, no memory effect, low self-discharge rate [95,96]. Compared to traditional lead-acid batteries and nickel chromium batteries, lithium-ion batteries have advantages of high energy density, long cycle life, high operating voltage, no memory effect, low self-discharge rate, and less pollution. Meanwhile, compared to various dye batteries, air batteries, and supercapacitor batteries, lithium-ion battery technology is significantly mature and cost-effective. These obvious advantages contribute to the widespread adoption of lithium-ion battery technology. Due to the high or low temperature of lithium-ion batteries, it can affect their normal operation. Fig. 13 displays the charging and discharging curve of battery with temperature changes [97]. The standard temperature range of the battery is 20–60 °C, if the temperature is too high, it may have a great impact on

the life and performance of battery, and more serious may lead to chain reactions such as thermal runaway [98]. Therefore, battery thermal management is an effective way to ensure the normal working of the battery.

The safety issue of lithium-ion batteries is actually their thermal safety, and the energy density, service life, discharge rate of batteries are greatly affected by temperature. At low temperatures, the battery capacity is low and the charging and discharging performance is poor. At high temperatures, the cycle life of batteries is shortened, and the chain exothermic reactions leads to thermal runaway and even serious accidents such as smoking, fire, and even explosions. Therefore, effective battery thermal management helps maintain the performance and ensure safe operation of the battery. The temperature may increase dramatically during use and trigger a self-inducing thermal reaction, eventually leading to thermal runaway or catastrophic damage [99]. Therefore, an efficient cooling system is a suitable solution for the thermal safety of batteries. PCMs absorb or release a large amount of latent heat during solid-liquid or solid-liquid phase transitions, thereby maintaining the operating temperature of the battery. In addition to the

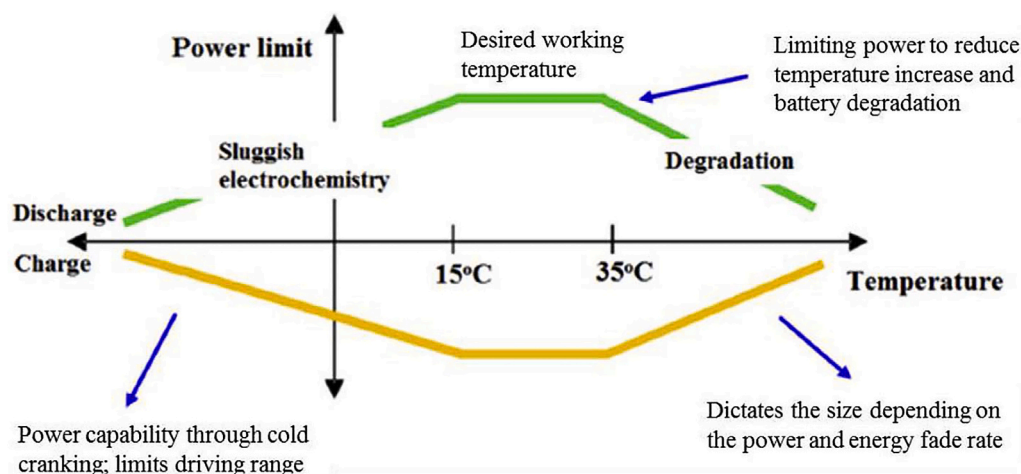


Fig. 13. The effect of temperature on battery power in charge and discharge process [97].

traditional air and liquid cooling, PCM can be used in battery thermal management. PCM absorbs the heat generated by the battery through its latent heat, and can help the battery pack within the desired temperature range [100]. There are several specific properties of PCMs for battery thermal management, such as large latent heat value, and high thermal conductivity, suitable phase change temperature.

An appropriate phase change temperature makes the battery in a suitable temperature environment, ensuring its performance and lifespan. When PCM is directly integrated into lithium-ion battery modules, many aspects need to be considered, such as a decrease in mass energy density of battery module, and an increase in the complexity of battery module structure. Meanwhile, we need to consider the density, latent heat, and thermal conductivity of PCMs, and hope to use fewer PCMs to meet the thermal management requirements of battery modules. The cost of PCM is also a key consideration in real applications. The increase in energy density of batteries over time is shown in Fig. 14. Continuous uncontrollable heat generation reaction may bring about unexpected heat accumulation [101]. Due to the anti-leakage and low subcooling of solid-solid PCMs, the application in battery thermal management has gradually emerged.

Lin et al. successfully achieved solid-solid phase change polymer by introducing a redox initiation system of BPO and DMPT. The phase change polymer demonstrated a suitable phase change point of 47.8 °C, high thermal conductivity of 2.33 W/(m·K) and latent heat of 99.6 J/g, compared with traditional composite PCM during 15 charge–discharge cycles in the battery module, it achieved a lower maximum temperature [102]. Wu et al. fabricated a conductive phase change composites (PCCs) by constructing polyurethane/graphite nanoplatelets networks by dual-encapsulation method. The hybrid networks endowed the PCCs with excellent mechanical strength and stable thermal properties. The cross-linked network alleviated the liquid leakage of PEG, and exhibited a latent heat of 163.5 J/g. Furthermore, it was proven to be suitable for battery thermal management for active preheating/passive cooling at a cold/hot ambient. Fig. 15 shows the design of PCC-based module for preheating and cooling functions of lithium-ion batteries [103].

Xiao et al. prepared a series of solid-solid PCMs with good thermal stability and adjustable phase change temperature (PCT) by free radical polymerization. This solid-solid PCMs with 3-dimensional main chain and alkyl side-chain showed excellent thermal stability and could

withstand the heat-erosion as high as 250 °C. Furthermore, the PCT region of solid-solid PCMs can be easily tailored in a wide range from 9.7 °C to 65.3 °C by adopting fitted alkyl acrylate monomers [104]. Yamagata et al. embedded a PCM into lithium-ion battery cell, and investigated its thermal and electrochemical behavior at vacuum environments. It was demonstrated to be applicable in the absence of active thermal control components [105]. Buonomo et al. implemented a simple electronics passive cooling device for lithium-ion batteries. A PCM material with metal foam was numerically investigated using the Ansys-Fluent code, and the melting times and maximum temperature were reported [106]. Liu et al. fabricated a thermo-responsive separator that absorbs thermal energy of the battery and improves its fire safety. Compared to the commercial polyolefin separator, it was coated with ceramic silica microcapsules. The PCM could store heat to retard the increase of battery temperature [107].

Wu et al. prepared carbon nanotube-urethane composite phase change materials (PU-CNT) with excellent thermal conductivity. In order to investigate the thermal management performance of PU-CNT, a thermal management test system was constructed using it as heat dissipation medium. The results indicated that the maximum temperature and maximum temperature difference of battery module were within 50 °C and 2 °C at 2C discharge rate [108]. Xia et al. successfully synthesized a series of novel polymeric solid-solid PCMs with lamellar structures by using PEG as the phase change ingredient, 4,4'-diphenylmethane diisocyanate as cross-linking agent, and graphene oxide (GO) as skeleton material to enhance the thermal conductivity of PCMs. The solid-solid PCMs displayed a regular and lamellar structure, which endowed the solid-solid PCMs with good thermal cycling stability and high thermal conductivity [8].

Yang et al. prepared a high thermal conductivity composite PCM with PEG/EG/HNT@AP through in situ chemical reduction and physical blending technology. Expanded graphite (EG) and silver nanoparticles (AgNPs) assembled into HNT (HNT@AP) played a synergistic role to construct an interconnected thermal network. As shown in Fig. 16. PEG as the phase change matrix and halloysite nanotube (HNT) as the support material can provide cross-linked network to prevent its leakage. PEG/EG/HNT@AP exhibited prominent thermal management performance in the battery module, and temperature was limited to below 60 °C at 3C discharge rate at an environment temperature of 35 °C

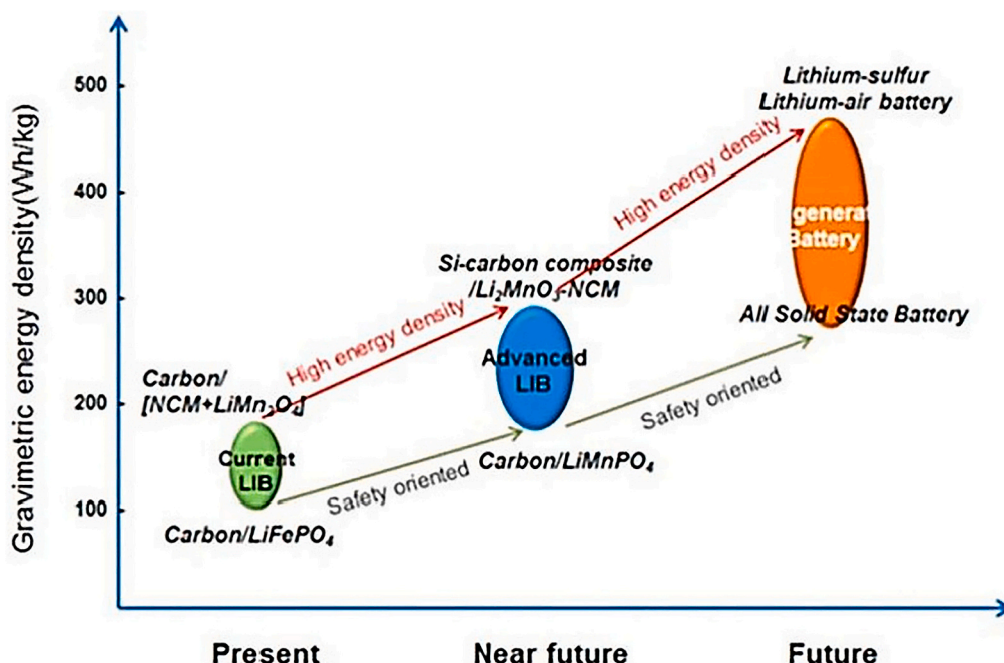


Fig. 14. Energy density trend of lithium-ion batteries over time [101].

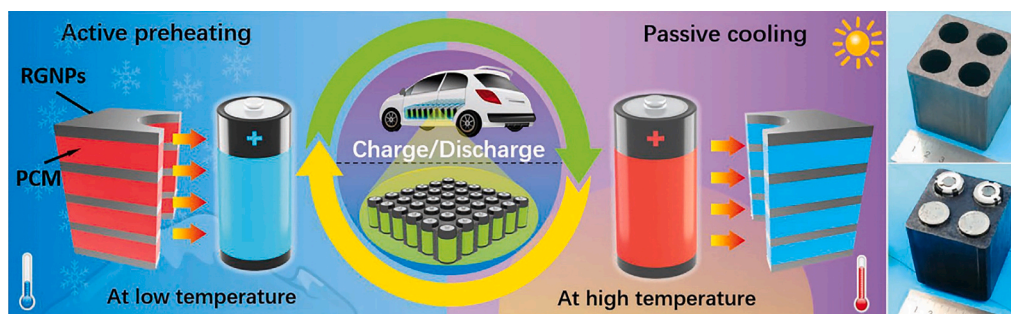


Fig. 15. Schematic of PCC-based module for battery thermal management [103].

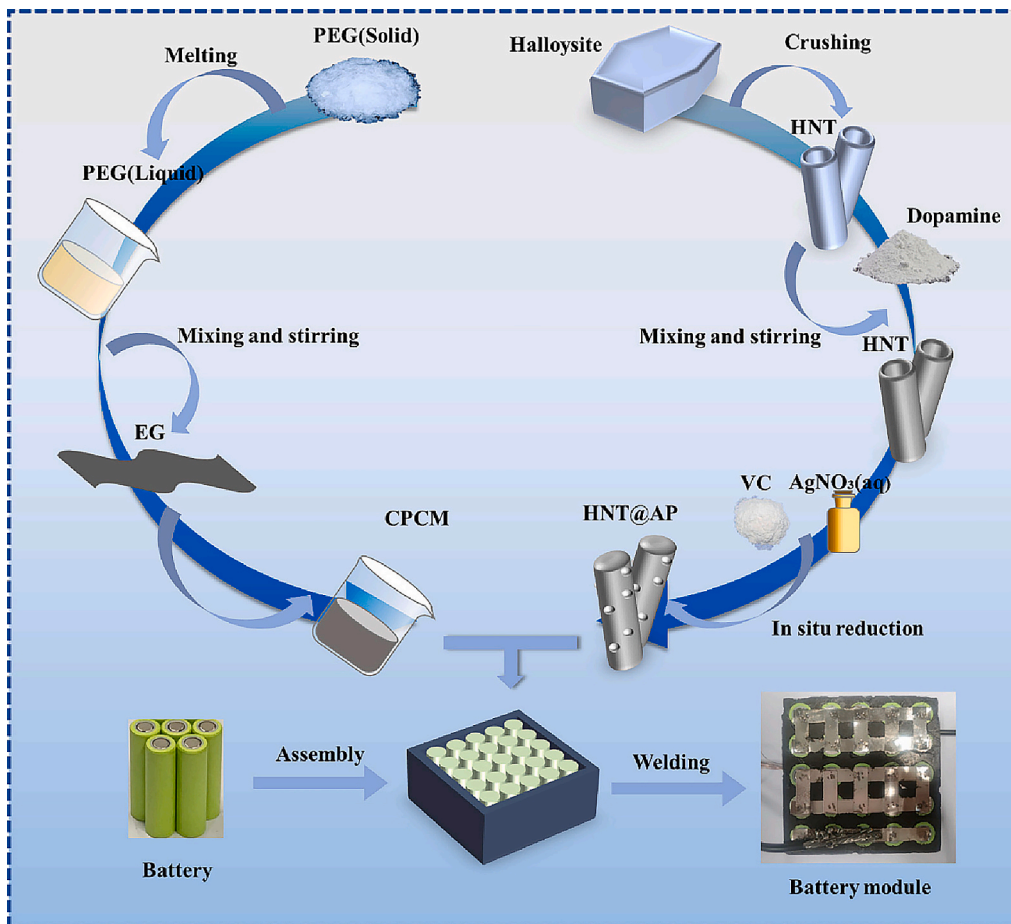


Fig. 16. The preparation of HNT@AP and the assembly procedure of CPCM for battery module [109].

[109].

Xiao et al. synthesized a kind of PoPCM in kilogram level with great thermal stability toward the practical application of battery thermal management. After incorporated with 6 wt% EG, the solid-solid PoPCM/EG-6 % composite obtained a high thermal conductivity of 2.33 W/(m·K), while the melting enthalpy decreased to 91.3 J/g. In consequence, the PoPCM/EG-6 % presented high cooling efficiency in battery module [110]. Xu et al. prepared a solid-solid PCM using PEG as phase change matrix and *N,N*-methylenebisacrylamide as the crosslinker. The effect of microcapsule-coated ammonium polyphosphate on flame retardancy and thermodynamic properties of composite PCM was investigated. And the battery thermal management system was designed and experiments were conducted [111].

Table 4 summarizes the composition and thermal performance of solid-solid PCMs for battery thermal management, and it can be seen

that the solid-solid PCMs synthesized by cross-linking polymerization have better thermal stability. In summary, the phase change temperature, latent heat capacity and thermal conductivity of PCMs play a very important role in battery thermal management. The appropriate phase change temperature can enable the heat generated by the lithium-ion battery in the process of charging and discharging to be absorbed by the PCM in time. The greater latent heat capacity, the more heat can be absorbed in the thermal management of the battery. The better the thermal conductivity, the faster the PCM can release the heat absorbed in the battery. There are a few studies of solid-solid PCMs for battery thermal management. However, its conductivity and enthalpy are low, so it needs to be greatly strengthened in these two aspects [112]. The current possible development areas of PCM battery thermal management are in electric vehicles, electric aerial vehicles, etc.

Starting from the thermal property requirements of PCMs in battery

Table 4

The composition and thermal performance of solid-solid PCMs.

Author	Ref.	Method	Composite PCM	Phase change temperature (°C)	Latent heat (J/g)	Thermal conductivity (W/(m·K))	Battery maximum temperature (°C)	Thermal stability (°C)
Lin et al.	[102]	Chemical grafting	OA + HDDA + BPO + EG	47.8	99.6	2.3	45.2	–
Wu et al.	[103]	Cross-linking polymerization	PEG + HDIB	43.4	120.6	3.1–27	48.6	280
Xiao et al.	[104]	Block copolymerization	TA/OA/DA/HA + HDDA+BPO + EG	47.6	92.6	–	52	250
Wu et al.	[108]	Cross-linking polymerization	PEG + CNT-OH + MDI	48.7	98.8	1.3	50.2	–
Yang et al.	[109]	Situ chemical reduction	PEG + HNT@AP + EG	12.53	103.65	1.15	52.2	312
Xiao et al.	[110]	Cross-linking polymerization	OA + HDDA+BPO + EG	43–50	98.8	1.31–2.26	45.8	–
Xu et al.	[111]	Cross-linking polymerization	PEG + MFAPP + MBA + acrylic acid	40–65	69.31–81.26	1.07	56.56	350

thermal management, solid-solid PCMs have advantages such as no leakage risk, no corrosion, and shape stability. Through the structural design of molecular chains and the regulation of composition formulas, researchers have developed a few solid-solid PCMs for battery thermal management. However, this research is still in a preliminary stage, and the latent heat and thermal conductivity of the synthesized solid-solid PCMs are relatively low, with less consideration in flame retardant performance. Further in-depth research is needed in the future.

4.2. Solar energy storage

Energy storage applications play a vital role in the utilization of solar technology. Solar energy storage includes two technologies, one is sensible heat storage and the other is latent heat storage [113,114]. Solid-liquid PCMs are currently commonly used in applications, but their leakage and corrosiveness will affect the application of phase change materials in solar energy storage. Therefore, solid-solid PCMs have been widely used in practice [115]. Solid-solid PCM is an ideal material in this regard due to its anti-leakage, non-toxicity, and non-corrosiveness.

Unlike traditional covalent crosslinking, dynamic cross-linking reactions improves the degradability of thermosetting solid-solid PCMs and meets the development needs of recycling. Molecular solar thermal compounds store photon energy in the form of strained chemical bonds during reversible photoisomerization process, and have become a new type of material that utilizes solar energy and releases stored energy as heat as needed. Wu et al. fabricated a novel dynamic cross-linked solid-solid PCMs through thermally reversible Diels-alder reaction by bonding PEG to polymeric skeleton, and the as-prepared DC-PCMs showed great potential in various fields such as solar energy storage, waste heat reutilization, smart textile, greenhouses, temperature adaptable buildings and so on [116].

Molecular solar thermal compounds are usually optical switches that undergo structural isomerization between the ground-state and metastable-state isomeric forms, and the energy difference between the two isomers is known as the isomerization energy, which is stored in the system. Thus phase-transition molecular solar thermal compounds exhibits large energy densities. Li et al. reported a series of adamantane-functionalized azobenzenes via reversible photoisomerization. They avoided the liquid encapsulation and showed good application prospects in solar energy storage devices [117]. Lu et al. synthesized PEG-based solid-solid PCMs using PEG and hexamethylene diisocyanate trimer by solvent-free approach [118]. The obtained solid-solid PCM displayed no leakage at 80 °C, which was higher than the melting temperature of PEG, its phase change temperatures, maximum latent heat, and relative enthalpy efficiency of were 65 °C, 136.8 J/g and 87.1 %, respectively. And the solid-solid PCMs exhibited excellent light-thermal conversion

ability and good thermal stability.

Alkan et al. synthesized polyurethane polymers using three different kinds of diisocyanate molecules and PEGs. The produced polyurethane polymers had typical solid-solid phase transition properties with high enthalpies of 179 J/g and transition points of 19–60 °C. These copolymers could be used in thermal energy storage and temperature-control fields [119]. Bing et al. developed a flexible, economic polymer supported PCMs with good heat transfer performance. The composite PCMs can be kept from leaking at 80 °C and retain the thermal properties and chemical structure after 50 melting and cooling cycles, and this study broadens an idea to realize high efficiency utilization in solar thermal systems [120].

Zheng et al. prepared a novel form-stable composite PCM for solar energy conversion and storage application. It exhibited an excellent light-to-thermal energy conversion efficiency as high as 97 % [121]. Zhou et al. synthesized polyurethane-based solid-solid PCMs from PEG and hexamethylene diisocyanate biuret, and the influence of halloysite nanotubes (HNTs) on this system was investigated. HNTs improved the solid-solid phase change properties. The system had no leakage even at the temperature that 40 °C higher above the melting temperature of PEG, and the latent heat increased from 86.8 to 118.7 J/g. Besides, heterogeneous nucleation effect could make up partial latent heat loss caused by crosslinking [122].

Gao et al. prepared a solid-solid polyurethane PCM by two-step solvent method using 4,4'-diphenylmethane diisocyanate as framework, hydroquinone bis (2-hydroxyethyl) ether as chain extender, and montmorillonite as nucleating agent [123], and the synthesis diagram is shown in Fig. 17(a). And the influence of montmorillonite on the phase transition property of PCM was investigated. When the addition of montmorillonite was 0.5 wt%, the phase change enthalpy of PCM increased from 95.46 to 106.8 J/g, a constant temperature platform appeared for about 300 s in heating and cooling stages, thus demonstrating excellent temperature control ability. Fig. 17(b) shows their heat transfer and storage capacity during the heating process [123].

Zhou et al. fabricated crosslinking polyurethane-based solid-solid PCMs with in situ reduced graphene oxide sheets by a facile solvothermal treatment. GO sheets could endow the solid-solid PCMs with outstanding form-stability above 100 °C due to the cooperative effect with hexamethylene diisocyanate biuret, and the light-thermal conversion efficiency was up to 78.7 % because of the simultaneous in situ reduction of GO. Moreover, an effect barrier formed by GO sheets enhanced the thermal stability of the system. The phase change enthalpy increased to 127.3 J/g from 86.8 J/g at 3.8 wt% GO addition for nucleation effects [124].

Mu et al. synthesized polyurethane-grafted reduced graphene oxide (rGO-PU) from PEG, 4,4'-diphenylmethane diisocyanate and rGO by in-situ polymerization, esterification, and reduction reaction. The thermal

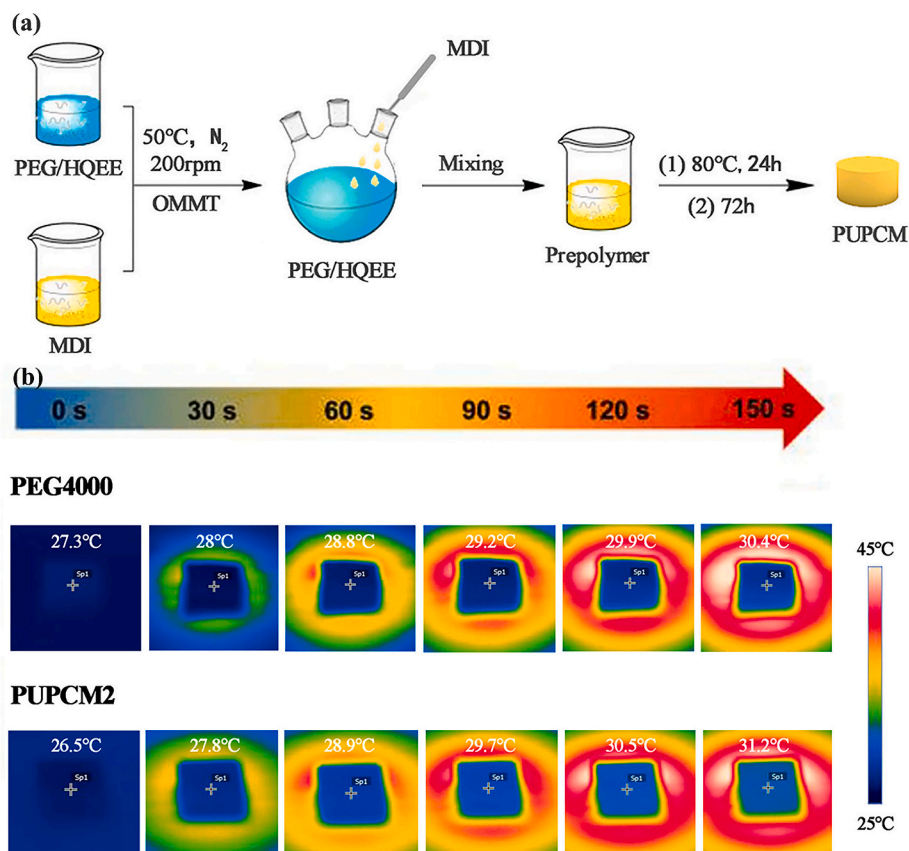


Fig. 17. (a) The schematic diagram of polyurethane preparation process, and (b) thermal transport evolution of PEG and PUPCM2 during the heating process [123].

conductivity of PU was substantially improved by rGO, the highest thermal conductivity of rGO-PU was 131.2 % higher than PU. The highest latent heats of melting and crystallization of rGO-PU-1 respectively reached to 138.7 J/g and 130.9 J/g [125]. Xiao et al. also developed a pentaglycerol-based composite PCM with the ability to convert and store solar and electric thermal energy [126].

There are a few applications of solid-solid PCMs in the field of solar energy. Although solid-solid PCMs have good thermal stability, their enthalpy and thermal conductivity are low. Most of the materials used are polyethylene glycol and paraffin, and although the enthalpy of the material itself is large, it is a solid-liquid phase change, and there is a leakage problem. PEG is generally combined with other hard segments to restrict its flow. Since low enthalpy and low thermal conductivity are two deficiencies of solid-solid phase change, additives are needed to supplement this aspect. The integration of solid-solid PCMs into solar energy systems, the problems of cycling stability and solar energy utilization efficiency. The thermal storage capacity of PCMs can be improved by adding stabilizers or solar absorbers.

4.3. Building energy conservation

Construction industry is developing rapidly around the world. According to statistics, buildings account for 35 % of the total energy consumption in the world, and generate 40 % of carbon dioxide emissions [127]. And most of the energy consumption is used for heating, air conditioning and other systems. Therefore, it is very important to develop effective and sustainable thermal management techniques. The PCMs applied in building energy conservation construction are mainly solid-liquid PCMs. The solid-liquid PCMs have the advantages of low raw material cost and moderate phase change temperature. The use of microcapsule and porous carrier adsorption can solve the leakage problem of PCMs, but the anti-leakage preparation process is complex

and the latent heat is greatly reduced. Compared with solid-liquid PCMs, solid-solid PCMs are directly processed and formed, leak-resistant, non-toxic, non-corrosive, no subcooling and phase separation phenomenon, which is very suitable to building energy conservation.

For more than 70 years, PCMs have increased the thermal mass of buildings through their large latent heat. Therefore, the development of thermal energy storage utilizing PCM has been one of the effective methods in building thermal management [128]. As illustrated in Fig. 18, the solidification of PCM happens during night time when cool ambient air flows through liquid PCM storage. Conversely, the melting process of PCM occurs during day time when hot air passes through solid PCM storage [129].

Solid-solid phase change has the advantages of anti-leakage performance compared with solid-liquid phase change, so it has received more attention in building energy conservation [130]. Tan et al. prepared form-stable PCMs utilizing PEG spherulite crystals as templates, and the cross-linked polymer as a supporting material. PCMs possessed a good thermally reversible reusability, good thermal stability, and multiple cycle stability. Fig. 19 exhibits a schematic diagram of recovery mechanism during reuse process. In addition, they also showed excellent cooling effect and insulation characteristics [131].

Jia et al. designed a heat storage asphalt pavement material and then fabricated asphalt mixtures. The synthesized polyurethane-based PCM displayed suitable phase change temperatures, desirable thermal energy storage capacity and good cycling stability, and the active cooling of asphalt pavement was achieved [132]. Fan et al. adopted a simple aqueous polymerization method to synthesize shape-stable cellulose nanocrystal based solid-solid PCM. The synthesized PCM had a melting temperature of 47.1 °C and phase change enthalpy was up to 82.3 J/g. Moreover, it showed excellent temperature regulation ability, and the temperature was adjusted within a comfortable range (15–30 °C) even if the ambient temperature changes quickly [133].

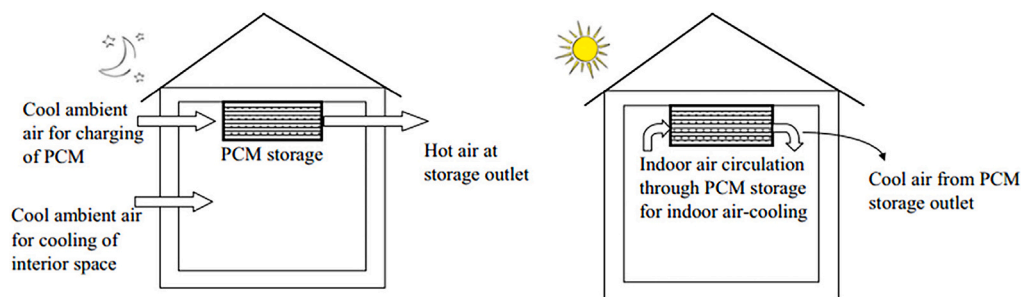


Fig. 18. Charging and discharging process of PCM storage [129].

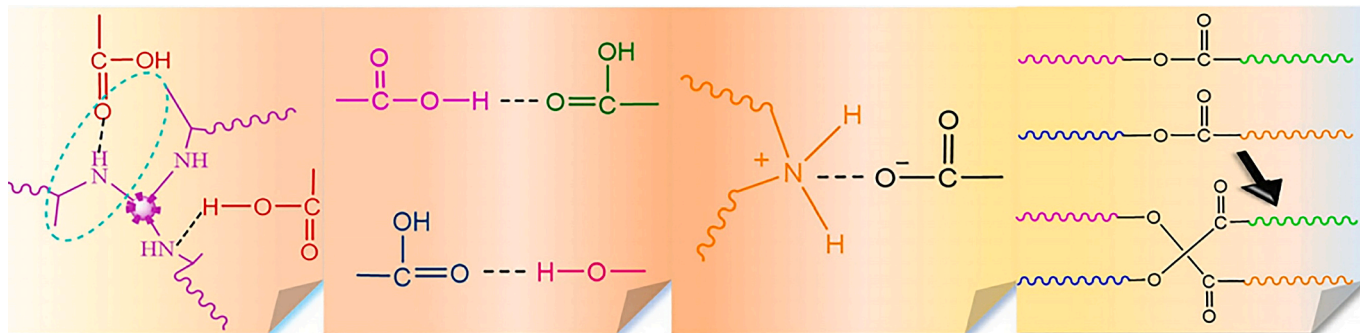


Fig. 19. The recovery mechanisms of thermally reversible cross-linked polymer [131].

Sha et al. dedicated to developing polyurethane PCMs suitable for cooling and utilization of asphalt pavement. The polyurethane PCMs with high isocyanate content could typically maintain high temperatures without liquid leakage [134]. Wang et al. prepared several polyurethane solid-solid PCM samples by a two-step condensation reaction. Considering the phase change temperature and thermal stability, PUPCM samples with appropriate molar ratios was used for asphalt modification. PCM can actively absorb and release heat to maintain the target temperature [135].

Harlé et al. successfully prepared a composite made of plaster and PCM. The amount and grain size of PCM showed an impact on mechanical resistance of the composite. The endothermic phase change of PCM delayed the rise in temperature of building material at high temperatures [136]. Khomein et al. designed PEG-based methacrylate polymers by utilizing a simple copolymerization strategy to tune their static tunability transition temperatures. The solid-solid PCMs displayed tunable transition temperatures ranged from 18 °C to 35 °C, suitable to building materials [137]. Zhang et al. prepared a new solid-solid PCM, which is mainly used in glass windows, and the latent heat of phase change and the thermal properties of optical properties of the materials are all used by glass windows to play a role in heat preservation in severe cold weather [138].

The application of solid-solid PCMs in building materials has been gradually welcomed in recent years. Because the requirements for enthalpy of building thermal management are not as high as those in other fields, the solid-solid PCMs are suitable due to their non-leakage, non-toxic, and non-corrosive properties. In addition, through the heat absorption or exothermic of PCMs during phase change process, the temperature of building is kept within the appropriate temperature range as far as possible to avoid large temperature fluctuations. In real applications, the durability and thermal properties of PCMs are crucial, such as good thermally reversible reusability, thermal stability, cycling stability, and suitable phase change temperatures.

4.4. Other thermal storage fields

PCMs can absorb or release heat at a constant temperature, and are widely used in thermal management of coatings, textiles, electronic device [139–141]. In most of literatures, the solid-liquid PCMs are widely researched [142]. However, solid-liquid PCMs have the notable shortcomings such as leakage and deformation [143]. Therefore, the solid-solid PCMs has increasingly got great attention.

Cui et al. reported a solid-solid coating with prominent form stability at high-temperatures because of the cross-linked network between PEG and MWCNTs. This coating showed obvious phase change property and high energy storage densities of 142.8 J/g [144]. Cao et al. fabricated a shape-stabilized solid-solid PCM of graphene oxide-grafted-poly (hexadecyl acrylate) through hexadecyl acrylate covalently bonding to graphene oxide nanosheets. The melting and freezing points were respectively 35.0 °C and 30.5 °C. The latent heats of melting and freezing were respectively 79 J/g and 77 J/g. The degradation temperature was approximately 214.4 °C, which was much higher than the working temperature region for energy storage applications [145].

Raj et al. synthesized manganese based layered perovskite solid-solid PCM with the addition of 5 wt% nano-encapsulated liquid gallium-indium metal alloy, and it exhibited high thermal stability, enhanced thermal conductivity and latent enthalpy, suitable for energy storage applications [146].

Liao et al. constructed a solid-solid PCM film with outstanding shape-stable property and super elastic, the PU single chain is displayed in Fig. 20(a). The phase change enthalpy and temperature of PU were adjusted through changing the molecular weight of PEG. It displayed long-term cycling stability after 700 thermal cycles and significant shape foldability and stretch ability, and the working mechanism of PU was shown in Fig. 20(b). Its elongation at break and enthalpy of PCM respectively reached to 420 % and 61.48 J/g [147].

Wei et al. synthesized polyurethane solid-solid PCM by prepolymer method, and then the asphalt modified by PCM was prepared. The addition of polyurethane solid-solid PCM improved the low temperature performance of asphalt. The coefficient of heat conductivity of modified

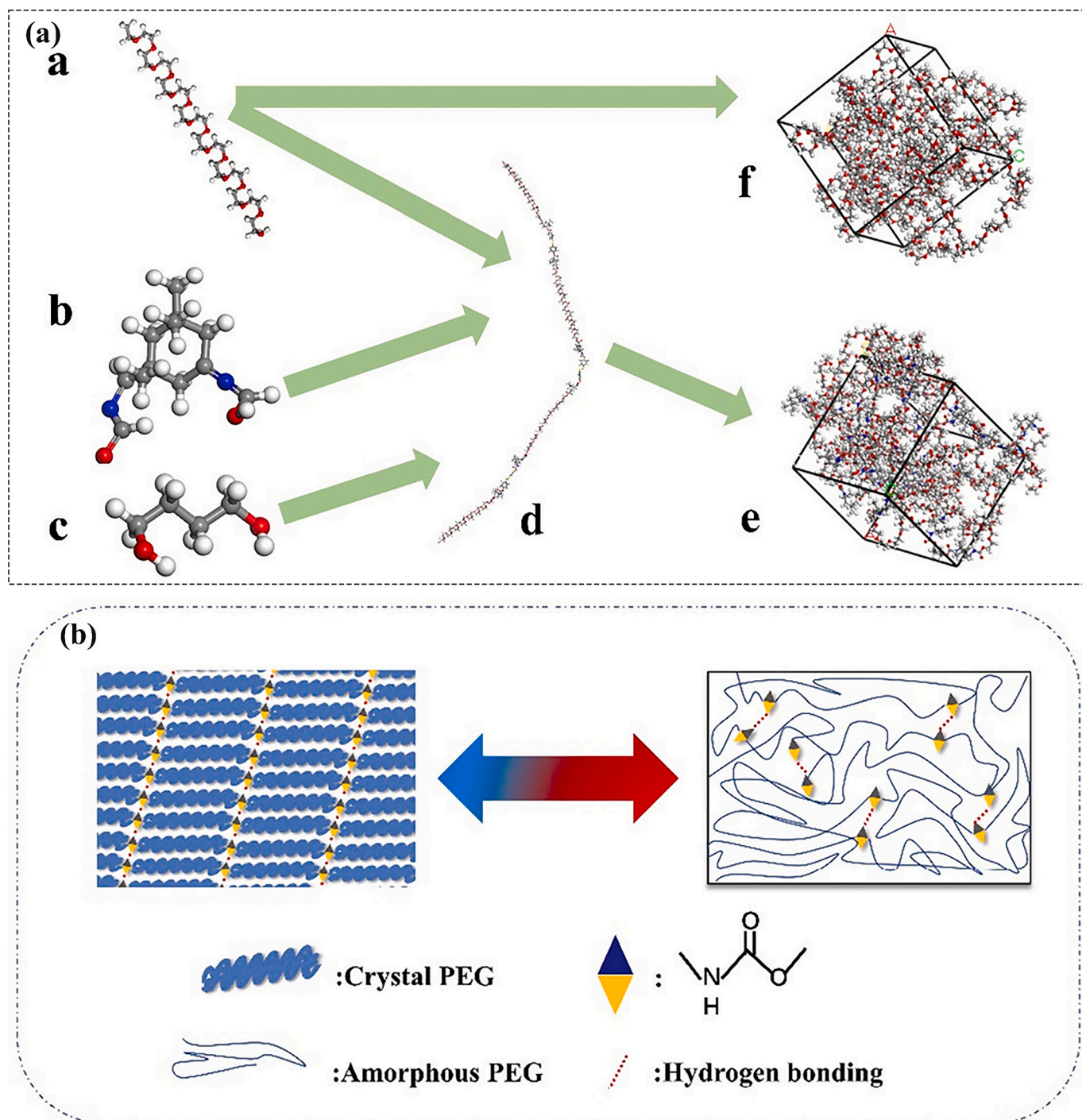


Fig. 20. (a) Simulated unit and cell construction process, and (b) the working mechanism of PU in the heating and cooling process [147].

asphalt gradually increased with increasing PCM content, due to the temperature control ability of modified asphalt [148]. Li et al. developed a series of novel solid-solid PCMs by employing the large latent heat and high thermal conductivity of these metallic Ni-Mn-Ti PCMs. Furthermore, they showed tunable phase change temperatures of 290–500 °C and superior thermal cycling stability [149]. Kong et al. reported a dynamic-covalent-crosslinking PCMs (V-PCMs) consisting of PEG as phase change ingredient, polyaryl polymethylene isocyanate as cross-linking points and disulfide as dynamic bonds. V-PCMs could repeatedly store and release heat during melting and crystallization process of PEG chains, exhibiting good thermal energy storage ability [150].

Wu et al. prepared a PEG solid-solid PCMs with good antistatic

properties by introducing ionic liquid form on the EPG macromolecular chain. The solid-solid PCMs showed good antistatic properties, and used for thermal energy storage and thermal management of electrical devices [151]. Chang et al. prepared a solid-solid PCMs based on PEGs of different molecular weights with an operating temperature range of -27.4 to 73.2 °C, which exhibited excellent thermal energy storage and thermal management capabilities under different climatic conditions [152].

Solid-solid PCMs have been widely used in other thermal storage fields due to their anti-leakage and non-deformation characteristics. However, the enthalpy of solid-solid PCMs needs to be improved. Besides, cycle stability is an important factor in practical applications. In

the current development of solid-solid phase change technology, the main bottleneck that currently restricts is reflected in the relatively low latent heat and thermal conductivity of solid-solid PCMs. The next focus of research is to achieve the preparation of solid-solid PCMs with high latent heat, high thermal conductivity, good thermal stability and chemical stability through effective molecular structure design and composition optimization.

5. Challenges and recommendations of solid-solid PCMs

5.1. Current challenges

In the utilization of solid-solid polymeric PCMs in real applications, they also have obvious shortcomings such as low latent heat, low thermal conductivity, and complex preparation process. These problems may be solved by adding thermal conductive agents, optimizing molecular chain structures, simplifying preparation processes, or using solvent-free methods. At present, raw materials of solid-solid PCMs are mainly extracted from petrochemical resources, which are non-renewable and do not conform to green and sustainable development. Then, the existing solid-solid PCMs mainly focus on their thermal storage performance, while there is relatively little research on multifunctional solid-solid PCMs. In practical applications, solid-solid PCMs are often required to have other functions at the same time, such as photo-thermal conversion, flame retardancy, flexibility, and the improvement of phase change enthalpy and thermal conductivity. Lastly, there is few research on the service life of solid-solid PCMs, and the aging mechanism of solid-solid PCMs during service should be further revealed.

5.2. Probable recommendations

First, more attention should be paid to design and preparation of recyclable solid-solid PCMs. To meet the requirements of sustainable development, it is a trend to prepare solid-solid PCMs through environmentally friendly process using renewable raw materials. It is necessary to consider the green design and manufacturing throughout the entire lifecycle. Thus, related researches should be conducted on raw materials, preparation process, biodegradability, and reprocessing properties. Secondly, the development of multifunctional solid-solid PCMs is worth further research. It should be focused on the multiple functions through selection of materials as well as additives, such as thermal energy conversion, fire safety, and electromagnetic shielding. Third, the impact of practical application scenarios on their performance should be investigated, such as high humidity, low temperature, high temperature, electromagnetic radiation, low air pressure, etc. Therefore, the thermal cycling and performance degradation behaviors of PCMs should be also explored, and these studies would provide experimental basis for development of solid-solid PCMs suitable for practical application scenarios. Lastly, in order to meet various specific application requirements, many factors need to be considered, such as high latent heat, high-thermal conductivity, various phase change temperature ranges, reasonable cost.

6. Conclusions

This review focuses on the application of solid-solid PCMs in various fields, from building energy saving to battery thermal management. The preparation of solid-solid PCMs was also discussed, including grafting, block, and cross-linking polymeric method. By designing molecular structures or filling with supporting materials, solid-liquid PCMs exhibit a macroscopic solid-solid phase change and still maintain their shape. On this basis, to make better use of solid-solid PCMs, the properties of enthalpy, thermal conductivity, and thermal stability of solid-solid PCMs need to be further improved. The research progress of solid-solid PCMs in various fields of thermal energy storage was also summarized. Although solid-solid PCMs face some difficulties in practical

use, the outstanding advantages of no phase separation and anti-leakage endow them good prospects for thermal energy storage applications.

CRediT authorship contribution statement

Maoyong Zhi: Writing – original draft, Methodology, Funding acquisition. **Shan Yue:** Writing – original draft, Methodology, Formal analysis. **Lingling Zheng:** Writing – original draft, Formal analysis. **Bingjian Su:** Resources, Data curation. **Ju Fu:** Project administration, Funding acquisition. **Qiang Sun:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Maoyong Zhi reports financial support was provided by Sichuan Province Science and Technology Support Program. Ju Fu reports financial support was provided by Sichuan Province Science and Technology Support Program. Maoyong Zhi reports financial support was provided by Fundamental Research Funds for the Central Universities. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was financially supported by supported by Sichuan Science and Technology Program (NO: 2022YFG0215, 2021SZY007), Fundamental Research Funds for the Central Universities (NO: X2023-2, XKJ2022-8).

References

- [1] Jiang, K.; Liao, G.; E, J.; Zhang, F.; Chen, J.; Leng, E. Thermal management technology of power lithium-ion batteries based on the phase transition of materials: a review. *J. Energ. Stor.* 2020, 32, 101816.
- [2] R. Huang, S. Zhang, P. Wang, Key areas and pathways for carbon emissions reduction in Beijing for the “Dual Carbon” targets, *J. Energy. Policy.* 164 (2022) 112873.
- [3] Y. Lv, X. Yang, G. Zhang, Durability of phase-change-material module and its relieving effect on battery deterioration during long-term cycles, *Appl. Therm. Eng.* 179 (2020) 115747.
- [4] G. Murali, G.S.N. Sravya, J. Jaya, A review on hybrid thermal management of battery packs and its cooling performance by enhanced PCM, *Renew. Sust. Energ. Rev.* 150 (2021) 111513.
- [5] A. Fallahi, G. Guldentops, M. Tao, Review on solid-solid phase change materials for thermal energy storage: molecular structure and thermal properties, *Appl. Therm. Eng.* 127 (2017) 1427–1441.
- [6] Y. Lv, G. Liu, G. Zhang, X. Yang, A novel thermal management structure using serpentine phase change material coupled with forced air convection for cylindrical battery modules, *J. Power Sources* 468 (2020) 228398.
- [7] V.V. Tyagi, K. Chopra, R.K. Sharma, A comprehensive review on phase change materials for heat storage applications: development, characterization, thermal and chemical stability, *Sol. Energy Mater. Sol. Cells* 234 (2022) 111392.
- [8] C.R. Raj, S. Suresh, R.R. Bhavsar, Recent developments in thermo-physical property enhancement and applications of solid solid phase change materials, *J. Therm. Anal. Calorim.* 139 (5) (2019) 3023–3049.
- [9] Singh, P.; Sharma, R. K.; Ansu, A. K. A comprehensive review on development of eutectic organic phase change materials and their composites for low and medium range thermal energy storage applications. *Sol. Energy Mater. Sol. Cells* 2021, 223, 110955.
- [10] Y. Xia, H. Zhang, P. Huang, Graphene-oxide-induced lamellar structures used to fabricate novel composite solid-solid phase change materials for thermal energy storage, *Chem. Eng. J.* 362 (2019) 909–920.
- [11] S. Lu, Q. Lin, Y. Liu, Study on thermal performance improvement technology of latent heat thermal energy storage for building heating, *Appl. Energy* 323 (2022) 119594.

- [12] N. Zhang, Y. Yuan, X. Cao, Latent heat thermal energy storage systems with solid-liquid phase change materials: a review, *Adv. Eng. Mater.* 20 (6) (2018) 1700753.
- [13] S.L. Zhou, Z.G. Zhang, X.M. Fang, Research progress of solid-solid phase change heat storage materials, *Chem. Ind. Prog.* 40 (03) (2021) 1371–1383.
- [14] M. Jia, A. Sha, W. Jiang, Developing a solid-solid phase change heat storage asphalt pavement material and its application as functional filler for cooling asphalt pavement, *Energ. Buildings* 285 (2023) 112935.
- [15] H. Pointner, W.D. Steinmann, Experimental demonstration of an active latent heat storage concept, *Appl. Energy* 168 (2016) 661–671.
- [16] M. Zhi, R. Fan, X. Yang, Recent research progress on phase change materials for thermal management of lithium-ion batteries, *J. Energ. Storage* 45 (2022) 103694.
- [17] Q. Lin, S. Wang, Z. Ma, Lattice Boltzmann simulation of flow and heat transfer evolution inside encapsulated phase change materials due to natural convection melting, *Chem. Eng. Sci.* 189 (2018) 154–164.
- [18] W. Aftab, A. Usman, J. Shi, K. Yuan, M. Qin, R. Zou, Phase change material-integrated latent heat storage systems for sustainable energy solutions, *Energy Environ. Sci.* 14 (8) (2021) 4268–4291.
- [19] S. Ramakrishnan, X. Wang, J. Sanjayan, Thermal performance of buildings integrated with phase change materials to reduce heat stress risks during extreme heatwave events, *Appl. Energy* 194 (2017) 410–421.
- [20] S.F. Ahmed, M. Khalid, W. Rashmi, Recent progress in solar thermal energy storage using nanomaterials, *Renew. Sust. Energ. Rev.* 67 (2017) 450–460.
- [21] M. Subramanian, A.T. Hoang, B. Kalidasan, S. Nizetic, J.M. Solomon, D. Balasubramanian, X.P. Nguyen, A technical review on composite phase change material based secondary assisted battery thermal management system for electric vehicles, *J. Clean. Prod.* 2021 (322) (2021) 129079.
- [22] W. Su, J. Darkwa, G. Kokogiannakis, Review of solid-liquid phase change materials and their encapsulation technologies, *Renew. Sust. Energ. Rev.* 48 (2015) 373–391.
- [23] Umair, M.; M. Zhang, Y.; Iqbal, K. Novel strategies and supporting materials applied to shape-stabilize organic phase change materials for thermal energy storage—a review, *Appl. Energy* 2019, 235, 846–873.
- [24] A. Kumar, S.K. Shukla, A review on thermal energy storage unit for solar thermal power plant application, *Energy Procedia* 74 (2015) 462–469.
- [25] Q. Sun, H. Zhang, J. Xue, Flexible phase change materials for thermal storage and temperature control, *Chem. Eng. J.* 353 (2018) 920–929.
- [26] Y. Zhang, J. Wang, J. Qiu, Ag-graphene/PEG composite phase change materials for enhancing solar-thermal energy conversion and storage capacity, *Appl. Energy* 237 (2019) 83–90.
- [27] Y. Kou, S. Wang, J. Lou, Thermal analysis and heat capacity study of polyethylene glycol (PEG) phase change materials for thermal energy storage applications, *J. Chem. Thermodyn.* 128 (2019) 259–274.
- [28] P. Pithappa, A. Kumar, S. Prakash, Chalcogenide phase change material for active terahertz photonics, *Adv. Mater.* 31 (12) (2019) 1808157.
- [29] I.A. Ramphal, M.E. Hagerman, Water-processable laponite/polyaniline/graphene oxide nanocomposites for energy applications, *Langmuir* 31 (4) (2015) 1505–1515.
- [30] H. Singh, A. Talekar, W.M. Chien, Continuous solid-state phase transitions in energy storage materials with orientational disorder—computational and experimental approach, *Energy* 91 (2015) 334–349.
- [31] X. Du, H. Wang, X. Cheng, Synthesis and thermal energy storage properties of a solid-solid phase change material with a novel comb-polyurethane block copolymer structure, *RSC Adv.* 6 (48) (2016) 42643–42648.
- [32] V. Busico, C. Carfagna, V. Salerno, The layer perovskites as thermal energy storage systems, *Sol. Energy* 24 (6) (1980) 575–579.
- [33] C. Wang, L. Feng, W. Li, Shape-stabilized phase change materials based on polyethylene glycol/porous carbon composite: the influence of the pore structure of the carbon materials, *Sol. Energ. Mat. Sol. C.* 105 (2012) 21–26.
- [34] M.H. Chung, J.C. Park, Development of PCM cool roof system to control urban heat island considering temperate climatic conditions, *Energ. Buildings* 116 (2016) 341–348.
- [35] L. Liu, H. Wang, X. Qi, L. Kong, J. Cui, X. Zhang, Shape-stabilized phase change materials based on poly (ethylene-graft-maleic anhydride)-g-alkyl alcohol comb-like polymers, *Sol. Energ. Mat. Sol. C.* 143 (2015) 21–28.
- [36] X. Du, H. Wang, X. Cheng, Z. Du, Synthesis and thermal energy storage properties of a solid-solid phase change material with a novel comb-polyurethane block copolymer structure, *RSC Adv.* 6 (48) (2016) 42643–42648.
- [37] W. Liu, Q. Lin, S. Chen, Microencapsulated phase change material through cellulose nanofibrils stabilized Pickering emulsion templating, *Adv. Compos. Hybrid Mater.* 6 (4) (2023) 149.
- [38] A. Sari, C. Alkan, A. Bicer, Synthesis and thermal properties of polystyrene-graft-PEG copolymers as new kinds of solid-solid phase change materials for thermal energy storage, *Mater. Chem. Phys.* 133 (1) (2012) 87–94.
- [39] J. Guo, P. Xie, X. Zhang, C. Yu, F. Guan, Y. Liu, Synthesis and characterization of graft copolymer of polyacrylonitrile-g-polyethylene glycol-maleic acid monoester macromonomer, *J. Appl. Polym. Sci.* 131 (2014) 87–94.
- [40] A. Sari, A. Biçer, C. Alkan, Thermal energy storage properties of polyethylene glycol grafted styrenic copolymer as novel solid-solid phase change materials, *Int. J. Energy Res.* 44 (5) (2020) 3976–3989.
- [41] N. Han, W. Zhang, X. Wang, X. Zhang, W. Li, W. Yu, Preparation and properties of shape-stabilized phase change material cellulose benzoate-g-polyoxyethylene (2) hexadecyl ether with potential for thermal energy storage, *Text. Res. J.* 89 (8) (2018) 1512–1521.
- [42] S. Mu, J. Guo, C. Yu, Y. Liu, Y. Gong, S. Zhang, L. Yang, S. Qi, A novel solid-solid phase change material based on poly(styrene-co-acrylonitrile) grafting with palmitic acid copolymers, *J. Macromol. Sci. A* 52 (8) (2015) 617–624.
- [43] A. Sari, A. Biçer, C. Alkan, Poly (styrene-co-maleic anhydride)-graft-fatty acids as novel solid-solid PCMs for thermal energy storage, *Polym. Eng. Sci.* 59 (s2) (2019) E337–E347.
- [44] N. Han, Z. Li, X. Zhang, W. Yu, X. Chen, D. Wang, J. Li, Synthesis and characterization of cellulose-g-polyoxyethylene (2) hexadecyl ether solid-solid phase change materials, *Cellulose* 23 (3) (2016) 1663–1674.
- [45] X. Huang, J. Guo, Y. Yang, S. Li, S. Chen, Y. Liu, S. Zhang, Synthesis and thermal properties of poly(acrylonitrile-co-allyl glycidyl ether)-graft-methoxypoly (ethylene glycol) copolymers as novel solid-solid phase-change materials for thermal energy storage, *J. Appl. Polym. Sci.* 135 (35) (2018) 46641.
- [46] A. Sari, A. Biçer, C. Alkan, Thermal energy storage characteristics of poly(styrene-co-maleic anhydride)-graft-PEG as polymeric solid-solid phase change materials, *Sol. Energ. Mater. Sol. C.* 161 (2017) 219–225.
- [47] C. Cheng, G. Cheng, F. Gong, Performance evaluation of asphalt mixture using polyethylene glycol polyacrylamide graft copolymer as solid-solid phase change materials, *Constr. Build. Mater.* 300 (2021) 124221.
- [48] X.H. Gu, B.Y. Xu, J.L. Zhou, S.W. Li, Studies on preparation and properties of PEG/MMT solid-solid phase change material, *Adv. Mater. Res.* 512 (2012) 1712–1715.
- [49] S.Y. Mu, J. Guo, Y.M. Gong, S. Zhang, Y. Yu, Synthesis and thermal properties of poly(styrene-co-acrylonitrile)-graft-polyethylene glycol copolymers as novel solid-solid phase change materials for thermal energy storage, *Chin. Chem. Lett.* 26 (11) (2015) 1364–1366.
- [50] J. Guo, H. Xiang, Q. Wang, Preparation of poly (decaglycerol-co-ethylene glycol) copolymer as phase change material, *Energ. Buildings* 48 (2012) 206–210.
- [51] Sari, A.; Alkan, C.; Biçer, A.; Karaipekli, A. Synthesis and thermal energy storage characteristics of polystyrene-graft-palmitic acid copolymers as solid-solid phase change materials. *Solar Energy Materials and Solar Cells*. 2011, 95(12), 3195–3201.
- [52] A. Sari, C. Alkan, Ö. Lafcı, Synthesis and thermal properties of poly (styrene-co-allyl alcohol)-graft-stearic acid copolymers as novel solid-solid PCMs for thermal energy storage, *Sol. Energy* 86 (9) (2012) 2282–2292.
- [53] Y. Li, R. Liu, Y. Huang, Synthesis and phase transition of cellulose-graft-poly (ethylene glycol) copolymers, *J. Appl. Polym. Sci.* 110 (3) (2008) 1797–1803.
- [54] M.Q. Wu, S. Wu, Y.F. Cai, R.Z. Wang, T.X. Li, Form-stable phase change composites: preparation, performance, and applications for thermal energy conversion, storage and management, *Energ. Stor. Mater.* 42 (2021) 380–417.
- [55] N. Radouane, A comprehensive review of composite phase change materials (cPCMs) for thermal management applications, including manufacturing processes, performance, and application, *Energies* 15 (21) (2022) 8271.
- [56] C.-L. Huang, L. Jiao, J.-J. Zhang, J.-B. Zeng, K.-K. Yang, Y.-Z. Wang, Poly (butylene succinate)-poly(ethylene glycol) multiblock copolymer: synthesis, structure, properties and shape memory performance, *Polym. Chem.* 3 (3) (2012) 800–808.
- [57] K. Pielichowska, M. Nowak, P. Szatkowski, B. Macherzyńska, The influence of chain extender on properties of polyurethane-based phase change materials modified with graphene, *Appl. Energy* 162 (2016) 1024–1033.
- [58] W. Wu, G. Ye, G. Zhang, Composite phase change material with room-temperature-flexibility for battery thermal management, *Chem. Eng. J.* 428 (2022) 131116.
- [59] C. Tian, J. Ning, Y. Yang, Super tough and stable solid-solid phase change material based on π - π stacking, *Chem. Eng. J.* 429 (2022) 132447.
- [60] L. Jiang, Y. Lei, X. Fu, H. Xu, S. Zhou, Y. Chen, J. Lei, Preparation of flexible solid-solid phase change materials with simultaneously thermal energy storage capability, reprocess ability and dual-actuated shape memory performance, *Polymer* 248 (2022) 124826.
- [61] D.F. Miranda, C. Versek, M.T. Tuominen, Cross-linked block copolymer/ionic liquid self-assembled blends for polymer gel electrolytes with high ionic conductivity and mechanical strength, *Macromolecules* 46 (23) (2013) 9313–9323.
- [62] C. Chen, W. Liu, Z. Wang, Novel form stable phase change materials based on the composites of polyethylene glycol/polymeric solid-solid phase change material, *Sol. Energ. Mater. Sol. C.* 134 (2015) 80–88.
- [63] R. Wang, Y. Xiao, J. Lei, A solid-solid phase change material based on dynamic ion cross-linking with reprocessability at room temperature, *Chem. Eng. J.* 390 (2020) 124586.
- [64] Y. Zhao, Z. Liu, Y. Wang, Q. Liu, J. Lei, C. Zhou, Water insensitive and solvent-free synthesis of biodegradable solid-solid phase change materials based on poly (ethylene glycol) for thermal energy storage, *Adv. Polym. Technol.* 37 (8) (2018) 3818–3830.
- [65] T. Harlé, G.T.M. Nguyen, B. Ledesert, Y. Mélinge, R.L. Hebert, Cross-linked polyurethane as solid-solid phase change material for low temperature thermal energy storage, *Thermochim. Acta* 685 (2020) 179191.
- [66] C. Lin, P. Ying, M. Huang, P. Zhang, T. Yang, G. Liu, J. Wu, V. Levchenko, Processable and recyclable polyurethane/HNTs @ Fe₃O₄ solid-solid phase change materials with excellent thermal conductivity for thermal energy storage, *Polym. Compos.* 42 (12) (2021) 6816–6826.
- [67] X. Du, L. Jin, S. Deng, M. Zhou, Z. Du, X. Cheng, H. Wang, Recyclable, self-healing, and flame-retardant solid-solid phase change materials based on thermally reversible cross-links for sustainable thermal energy storage, *ACS Appl. Mater. Interfaces* 13 (36) (2021) 42991–43001.

- [68] L. Huang, Y. Yang, D. Yuan, X. Cai, Solid-solid phase-change materials with excellent mechanical property and solid state plasticity based on dynamic urethane bonds for thermal energy storage, *J. Energ. Stor.* 36 (2021) 102343.
- [69] C. Chen, W. Liu, Z. Wang, K. Peng, W. Pan, Q. Xie, Novel form stable phase change materials based on the composites of polyethylene glycol/polymeric solid-solid phase change material, *Sol. Energ. Mater. Sol. C.* 134 (2015) 80–88.
- [70] G.F. Wang, D.Y. Li, G.L. Pei, Preparation and characterization of a novel polyurethane solid-solid phase change materials, *Adv. Mater. Res.* 785 (2013) 613–617.
- [71] X. Fu, L. Jiang, Y. Wang, Y. Xiao, B. Wu, W. Kong, J. Lei, Reprocessable, biodegradable polyester-based solid-solid phase change materials networks from dynamic ionic crosslinking with high latent heat capability, *J. Clean. Prod.* 297 (2021) 126630.
- [72] P. Zare, N. Perera, J. Lahr, R. Hasan, Solid-liquid phase change materials for the battery thermal management systems in electric vehicles and hybrid electric vehicles – a systematic review, *J. Energ. Stor.* 52 (2022) 105026.
- [73] R.K. Sharma, P. Ganesan, V.V. Tyagi, Developments in organic solid-liquid phase change materials and their applications in thermal energy storage, *Energ. Convers. Manage.* 95 (2015) 193–228.
- [74] F. Wang, P. Zhang, Y. Mou, M. Kang, M. Liu, L. Song, A. Lu, J. Rong, Synthesis of the polyethylene glycol solid-solid phase change materials with a functionalized graphene oxide for thermal energy storage, *Polym. Test.* 63 (2017) 494–504.
- [75] A.O. Elsayed, Numerical study on performance enhancement of solid-solid phase change materials by using multi-nanoparticles mixtures, *J. Energ. Stor.* 4 (2015) 106–112.
- [76] R. Luo, H. Lei, F. Shao, L. Wang, H. Xie, W. Yu, Graphene-pentaerythritol solid-solid phase change composites with high photothermal conversion and thermal conductivity, *Sol. Energy* 241 (2022) 54–62.
- [77] D. Mani, M.K. Saranprabhu, K.S. Rajan, Intensification of thermal energy storage using copper-pentaerythritol nanocomposites for renewable energy utilization, *Renew. Energy* 163 (2021) 625–634.
- [78] C. Zhang, Z. Shi, A. Li, Y.F. Zhang, RGO-coated polyurethane foam/segmented polyurethane composites as solid-solid phase change thermal interface material, *Polymers-Basel*. 12 (12) (2020) 3004.
- [79] C.Y. Zhao, Y.B. Tao, Y.S. Yu, Molecular dynamics simulation of nanoparticle effect on melting enthalpy of paraffin phase change material, *Int. J. Heat Mass Transf.* 150 (2020) 119382.
- [80] L. Yanshan, W. Shujun, L. Hongyan, M. Fanbin, M. Huanqing, Z. Wangang, Preparation and characterization of melamine/formaldehyde/polyethylene glycol crosslinking copolymers as solid-solid phase change materials, *Sol. Energ. Mater. Sol. C.* 127 (2014) 92–97.
- [81] T. Li, H. Pan, L. Xu, Shape-stabilized phase change material with high phase change enthalpy made of PEG compounded with lignin-based carbon, *Int. J. Biol. Macromol.* 213 (2022) 134–144.
- [82] C. Alkan, E. Günther, S. Hiebler, Ö.F. Ensari, D. Kahraman, Polyurethanes as solid-solid phase change materials for thermal energy storage, *Sol. Energy* 86 (6) (2012) 1761–1769.
- [83] Y. Zhou, X. Wang, X. Liu, D. Sheng, F. Ji, L. Dong, S. Xu, H. Wu, Y. Yang, Multifunctional ZnO/polyurethane-based solid-solid phase change materials with graphene aerogel, *Sol. Energ. Mater. Sol. C.* 193 (2019) 13–21.
- [84] X. Du, H. Wang, Z. Du, X. Cheng, Synthesis and thermal properties of novel solid-solid phase change materials with comb-polyurethane block copolymer structure for thermal energy storage, *Thermochim. Acta* 651 (2017) 58–64.
- [85] J.L. Dauvergne, A. Serrano, E.P. Del Barrio, Fast estimation of the enthalpy-temperature function of phase change materials, *Exp. Thermal Fluid Sci.* 122 (2021) 110317.
- [86] Y. Zhang, M.M. Umair, S. Zhang, Phase change materials for electron-triggered energy conversion and storage: a review, *J. Mater. Chem. A* 7 (39) (2019) 22218–22228.
- [87] J. Shi, M. Qin, W. Aftab, Flexible phase change materials for thermal energy storage, *Energ. Stor. Mater.* 41 (2021) 321–342.
- [88] C.R. Raj, S. Suresh, R.R. Bhavsar, Manganese-based layered perovskite solid-solid phase change material: synthesis, characterization and thermal stability study, *Mech. Mater.* 135 (2019) 88–97.
- [89] W. Kong, X. Fu, Z. Liu, C. Zhou, J. Lei, A facile synthesis of solid-solid phase change material for thermal energy storage, *Appl. Therm. Eng.* 117 (2017) 622–628.
- [90] Y. Yang, W. Kong, X. Cai, Solvent-free preparation and performance of novel xylitol based solid-solid phase change materials for thermal energy storage, *Energ. Buildings* 158 (2018) 37–42.
- [91] Z. Liu, B. Wu, X. Fu, P. Yan, Y. Yuan, C. Zhou, J. Lei, Two components based polyethylene glycol/thermosetting solid-solid phase change material composites as novel form stable phase change materials for flexible thermal energy storage application, *Sol. Energ. Mater. Sol. C.* 170 (2017) 197–204.
- [92] C. Chen, W. Liu, H. Wang, K. Peng, Synthesis and performances of novel solid-solid phase change materials with hexahydroxy compounds for thermal energy storage, *Appl. Energy* 152 (2015) 198–206.
- [93] M.K. Rathod, J. Banerjee, Thermal stability of phase change materials used in latent heat energy storage systems: a review, *Renew. Sust. Energ. Rev.* 18 (2013) 246–258.
- [94] C.R. Raj, S. Suresh, R.R. Bhavsar, V.K. Singh, Recent developments in thermo-physical property enhancement and applications of solid solid phase change materials: a review, *J. Therm. Anal. Calorim.* 139 (2020) 3023–3049.
- [95] D. Ren, X. Feng, L. Lu, An electrochemical-thermal coupled overcharge-to-thermal-runaway model for lithium ion battery, *J. Power Sources* 364 (2017) 328–340.
- [96] X. Feng, M. Ouyang, X. Liu, Thermal runaway mechanism of lithium ion battery for electric vehicles: a review, *Energ. Stor. Mater.* 10 (2018) 246–267.
- [97] L. Ianniciello, P.H. Biwolé, P. Achard, Electric vehicles batteries thermal management systems employing phase change materials, *J. Power Sources* 378 (2018) 383–403.
- [98] A. Belgibayeva, A. Rakhmetova, M. Rakhmatyzy, Lithium-ion batteries for low-temperature applications: limiting factors and solutions, *J. Power Sources* 557 (2023) 232550.
- [99] L. Kong, C. Li, J. Jiang, Li-ion battery fire hazards and safety strategies, *Energies* 11 (9) (2018) 2191.
- [100] G. Ye, G. Zhang, L. Jiang, X. Yang, Temperature control of battery modules through composite phase change materials with dual operating temperature regions, *Chem. Eng. J.* 449 (2022) 137733.
- [101] A. Mitra, R. Kumar, D.K. Singh, Z. Said, Advances in the improvement of thermal-conductivity of phase change material-based lithium-ion battery thermal management systems: an updated review, *J. Energ. Stor.* 53 (2022) 105195.
- [102] X. Lin, D. Shao, L. Jiang, X. Dong, G. Zhang, C. Xiao, X. Yang, Low-temperature preparation of solid-solid phase change polymer for thermal management modules, *Chem. Eng. Sci.* 246 (2021) 116985.
- [103] M. Wu, T. Li, P. Wang, S. Wu, R. Wang, J. Lin, Dual-encapsulated highly conductive and liquid-free phase change composites enabled by polyurethane/graphite nanoplatelets hybrid networks for efficient energy storage and thermal management, *Small* 18 (9) (2022) e2105647.
- [104] C. Xiao, S. Li, S. Zhu, G. Zhang, X. Yang, Series of solid-solid phase change materials with ultra-high thermal stability and controllable phase change temperature: kilogram-level preparation and application investigation, *J. Energ. Stor.* 36 (2021) 102405.
- [105] M.R. Yamagata, Y. Wakita, Y. Tsuruda, K. Miyata, Feasibility study of low-temperature operable electric power supply for CubeSats using passive thermal control with VO₂-based solid-solid phase change material, *Thermal Science and Engineering Progress*. 37 (2023) 101601.
- [106] B. Buonomo, D. Ercole, O. Manca, Thermal cooling behaviors of lithium-ion batteries by metal foam with phase change materials, *Energy Procedia* 148 (2018) 1175–1182.
- [107] Z. Liu, Y. Peng, T. Meng, L. Yu, S. Wang, X. Hu, Thermal-triggered fire-extinguishing separators by phase change materials for high-safety lithium-ion batteries, *Energ. Stor. Mater.* 47 (2022) 445–452.
- [108] T. Wu, C. Wang, Y. Hu, Flexible solid-solid phase change materials with high stability for thermal management, *Int. J. Heat Mass Transf.* 211 (2023) 124202.
- [109] W. Yang, R. Lin, X. Li, High thermal conductive and anti-leakage composite phase change material with halloysite nanotube for battery thermal management system, *J. Energ. Stor.* 66 (2023) 107372.
- [110] C. Xiao, G. Zhang, Z. Li, X. Yang, Custom design of solid-solid phase change material with ultra-high thermal stability for battery thermal management, *J. Mater. Chem. A* 8 (29) (2020) 14624–14633.
- [111] Z. Xu, W. Chen, T. Wu, Thermal management system study of flame retardant solid-solid phase change material battery, *Surf. Interfaces*. 36 (2023) 102558.
- [112] R. Naveenkumar, M. Ravichandran, V. Mohanavel, A. Karthick, L. Aswin, S.S. H. Priyanka, S.K. Kumar, S.P. Kumar, Review on phase change materials for solar energy storage applications, *Environ. Sci. Pollut. Res. Int.* 29 (7) (2022) 9491–9532.
- [113] S. Harikrishnan, S. Imran Hussain, A. Devaraju, P. Sivasamy, S. Kalaiselvam, Improved performance of a newly prepared nano-enhanced phase change material for solar energy storage, *J. Mech. Sci. Technol.* 31 (10) (2017) 4903–4910.
- [114] A.A. Abduljalil, A. Al, S.B. Mat, Review of thermal energy storage for air conditioning systems, *Renew. Sust. Energ. Rev.* 16 (2012) 5802–5819.
- [115] Z. Zhou, Z. Zhang, J. Zuo, K. Huang, L. Zhang, Phase change materials for solar thermal energy storage in residential buildings in cold climate, *Renew. Sust. Energ. Rev.* 48 (2015) 692–703.
- [116] Wu, B.; Wang, Y.; Liu, Z.; Liu, Y.; Fu, X.; Kong, W.; Jiang, L.; Yuan, Y.; Zhang, X.; Lei, J. Thermally reliable, recyclable and malleable solid-solid phase-change materials through the classical Diels-Alder reaction for sustainable thermal energy storage. *J. Mater. Chem. A* 2019, 7(38), 21802–21811.
- [117] X. Li, S. Cho, G.G. Han, Light-responsive solid-solid phase change materials for photon and thermal energy storage, *ACS Mater. Au.* 3 (1) (2022) 37–42.
- [118] Lu, X.; Fang, C.; Sheng, X.; Zhang, L.; Qu, J. One-step and solvent-free synthesis of polyethylene glycol-based polyurethane as solid-solid phase change materials for solar thermal energy storage. *Ind. Eng. Chem. Res.* 2019, 58(8), 3024–3032.
- [119] C. Alkan, E. Günther, S. Hiebler, Ö.F. Ensari, D. Kahraman, Polyurethanes as solid-solid phase change materials for thermal energy storage, *Sol. Energy* 86 (6) (2012) 1761–1769.
- [120] N. Bing, J. Yang, H. Gao, Unsaturated polyester resin supported form-stable phase change materials with enhanced thermal conductivity for solar energy storage and conversion, *Renew. Energy* 173 (2021) 926–933.
- [121] X. Zheng, X. Gao, Z. Huang, Form-stable paraffin/graphene aerogel/copper foam composite phase change material for solar energy conversion and storage, *Sol. Energ. Mater. Sol. C.* 226 (2021) 111083.
- [122] Y. Zhou, D. Sheng, X. Liu, C. Lin, F. Ji, L. Dong, S. Xu, Y. Yang, Synthesis and properties of crosslinking halloysite nanotubes/polyurethane-based solid-solid phase change materials, *Sol. Energ. Mater. Sol. C.* 174 (2018) 84–93.
- [123] N. Gao, T. Tang, H. Xiang, W. Zhang, Y. Li, C. Yang, X. Liu, Preparation and structure-properties of crosslinking organic montmorillonite/polyurethane as solid-solid phase change materials for thermal energy storage, *Sol. Energ. Mater. Sol. C.* 244 (2022) 111831.

- [124] Y. Zhou, X. Liu, D. Sheng, C. Lin, F. Ji, L. Dong, S. Xu, H. Wu, Y. Yang, Polyurethane-based solid-solid phase change materials with in situ reduced graphene oxide for light-thermal energy conversion and storage, *Chem. Eng. J.* 338 (2018) 117–125.
- [125] B. Mu, M. Li, Fabrication and characterization of polyurethane-grafted reduced graphene oxide as solid-solid phase change materials for solar energy conversion and storage, *Sol. Energy* 188 (2019) 230–238.
- [126] Q. Xiao, Y. Xu, X. Li, H. Tang, H. Li, L. Zhang, Enhanced solar-thermal and electro-thermal storage performance of solid-solid composite phase change material, *Composites Communications* 45 (2024) 101818.
- [127] N. Atmaca, A. Atmaca, A.İ. Özçetin, The impacts of restoration and reconstruction of a heritage building on life cycle energy consumption and related carbon dioxide emissions, *Energ. Buildings* 253 (2021) 111507.
- [128] H. Akeiber, P. Nejat, M.Z.A. Majid, A review on phase change material (PCM) for sustainable passive cooling in building envelopes, *Renew. Sust. Energ. Rev.* 60 (2016) 1470–1497.
- [129] A. Waqas, Din Ud, Z., Phase change material (PCM) storage for free cooling of buildings—a review, *Renew. Sust. Energ. Rev.* 18 (2013) 607–625.
- [130] Y. Wang, B. Wu, Y. Zhao, Q. Liu, J. Lei, Study on a novel solid–solid phase change materials: solvent-free preparation, thermal properties and phase separation behaviors, *J. Therm. Anal. Calorim.* 141 (4) (2020) 1305–1315.
- [131] Y. Tan, Y. Xiao, R. Chen, C. Zhou, L. Wang, Y. Liu, D. Li, High latent heat and recyclable form-stable phase change materials prepared via a facile self-template method, *Chem. Eng. J.* 396 (2020) 125265.
- [132] M. Jia, A. Sha, W. Jiang, X. Li, W. Jiao, Developing a solid–solid phase change heat storage asphalt pavement material and its application as functional filler for cooling asphalt pavement, *Energ. Buildings* 285 (2023) 112935.
- [133] X. Fan, Y. Guan, Y. Li, H.-Y. Yu, J. Marek, D. Wang, J. Militky, Z.-Y. Zou, J. Yao, Shape-stabilized cellulose nanocrystal-based phase-change materials for energy storage, *ACS Appl. Nano Mater.* 3 (2) (2020) 1741–1748.
- [134] A. Sha, J. Zhang, M. Jia, W. Jiang, W. Jiao, Development of polyurethane-based solid-solid phase change materials for cooling asphalt pavements, *Energ. Buildings* 259 (2022) 111873.
- [135] X. Wang, B. Ma, K. Wei, W. Si, X. Kang, Y. Fang, X. Zhou, Thermal storage properties of polyurethane solid-solid phase-change material with low phase-change temperature and its effects on performance of asphalt binders, *J. Energ. Stor.* 55 (2022) 105686.
- [136] T. Harlé, R.L. Hebert, G.T.M. Nguyen, B.A. Ledéser, A composite of cross-linked polyurethane as solid–solid phase change material and plaster for building application, *Energ. Buildings* 262 (2022) 111945.
- [137] P. Khomein, A. Nallapaneni, J. Lau, D. Lilley, C. Zhu, S. Kaur, R. Prasher, G. Liu, Random copolymer of poly(polyethylene glycol methyl ether)methacrylate as tunable transition temperature solid-solid phase change material for thermal energy storage, *Sol. Energ. Mat. Sol. C.* 225 (2021) 111030.
- [138] C. Zhang, R. Yang, Y. Lu, M. Arici, Y. Ma, X. Yang, Z. Qi, D. Li, Parametric research on thermal and optical properties of solid-solid phase change material packaged in glazing windows, *Journal of Energy Storage* 83 (2024) 110562.
- [139] D. Liu, C. Lei, K. Wu, Q. Fu, A multidirectionally thermoconductive phase change material enables high and durable electricity via real-environment solar–thermal–electric conversion, *ACS Nano* 14 (11) (2020) 15738–15747.
- [140] W. Aftab, X. Huang, W. Wu, Z. Liang, A. Mahmood, R. Zou, Nanoconfined phase change materials for thermal energy applications, *Energy Environ. Sci.* 11 (6) (2018) 1392–1424.
- [141] H. Liu, J. Tang, L. Dong, H. Wang, T. Xu, W. Gao, W. Feng, Optically triggered synchronous heat release of phase-change enthalpy and photo-thermal energy in phase-change materials at low temperatures, *Adv. Funct. Mater.* 31 (6) (2021) 2008496.
- [142] H. Nazir, M. Batool, F.J.B. Osorio, M. Isaza-Ruiz, X. Xu, K. Vignarooban, A. M. Kannan, Recent developments in phase change materials for energy storage applications: a review, *Int. J. Heat Mass Transf.* 129 (2019) 491–523.
- [143] X. Huang, X. Chen, A. Li, D. Atinafu, H. Gao, W. Dong, G. Wang, Shape-stabilized phase change materials based on porous supports for thermal energy storage applications, *Chem. Eng. J.* 356 (2019) 641–661.
- [144] J. Cui, W. Li, Y. Wang, H. Yu, X. Feng, Z. Lou, W. Shan, Y. Xiong, Ultra-stable phase change coatings by self-cross-linkable reactive poly(ethylene glycol) and MWCNTs, *Adv. Funct. Mater.* 32 (10) (2021) 2108000.
- [145] R. Cao, H. Liu, S. Chen, D. Pei, J. Miao, X. Zhang, Fabrication and properties of graphene oxide-grafted-poly(hexadecyl acrylate) as a solid-solid phase change material, *Compos. Sci. Technol.* 149 (2017) 262–268.
- [146] C.R. Raj, S. Suresh, V.K. Singh, R.R. Bhavsar, M. Chandrasekar, V. Archita, Life cycle assessment of nanoalloy enhanced layered perovskite solid-solid phase change material till 10000 thermal cycles for energy storage applications, *J. Energ. Stor.* 35 (2021) 102220.
- [147] Y. Liao, J. Li, S. Li, X. Yang, Super-elastic and shape-stable solid-solid phase change materials for thermal management of electronics, *J. Energ. Stor.* (2022) 52.
- [148] K. Wei, X. Wang, B. Ma, W. Shi, S. Duan, F. Liu, Study on rheological properties and phase-change temperature control of asphalt modified by polyurethane solid–solid phase change material, *Sol. Energy* 194 (2019) 893–902.
- [149] S. Li, L. He, H. Lu, J. Hao, D. Wang, F. Shen, D. Cong, Ultrahigh-performance solid-solid phase change material for efficient, high-temperature thermal energy storage, *Acta Mater.* 249 (2023) 118852.
- [150] W. Kong, Y. Yang, A. Yuan, L. Jiang, X. Fu, Y. Wang, H. Xu, Z. Liu, J. Lei, Processable and recyclable crosslinking solid-solid phase change materials based on dynamic disulfide covalent adaptable networks for thermal energy storage, *Energy* 232 (2021).
- [151] S. Wu, Y. Zhang, M. Deng, G. Liang, H. Ju, Y. Xiao, An intrinsic antistatic polyethylene glycol-based solid–solid phase change material for thermal energy storage and thermal management, *Polym. Adv. Technol.* 35 (3) (2024) e6347.
- [152] J. Chang, H. Lu, Y. Xiao, S. Wu, Y. Xiao, Y. Wang, A solid–solid phase change material with wide service temperature range based on polyethylene glycols with different molecular weights, *J. Appl. Polym. Sci.* 141 (7) (2024) e54950.